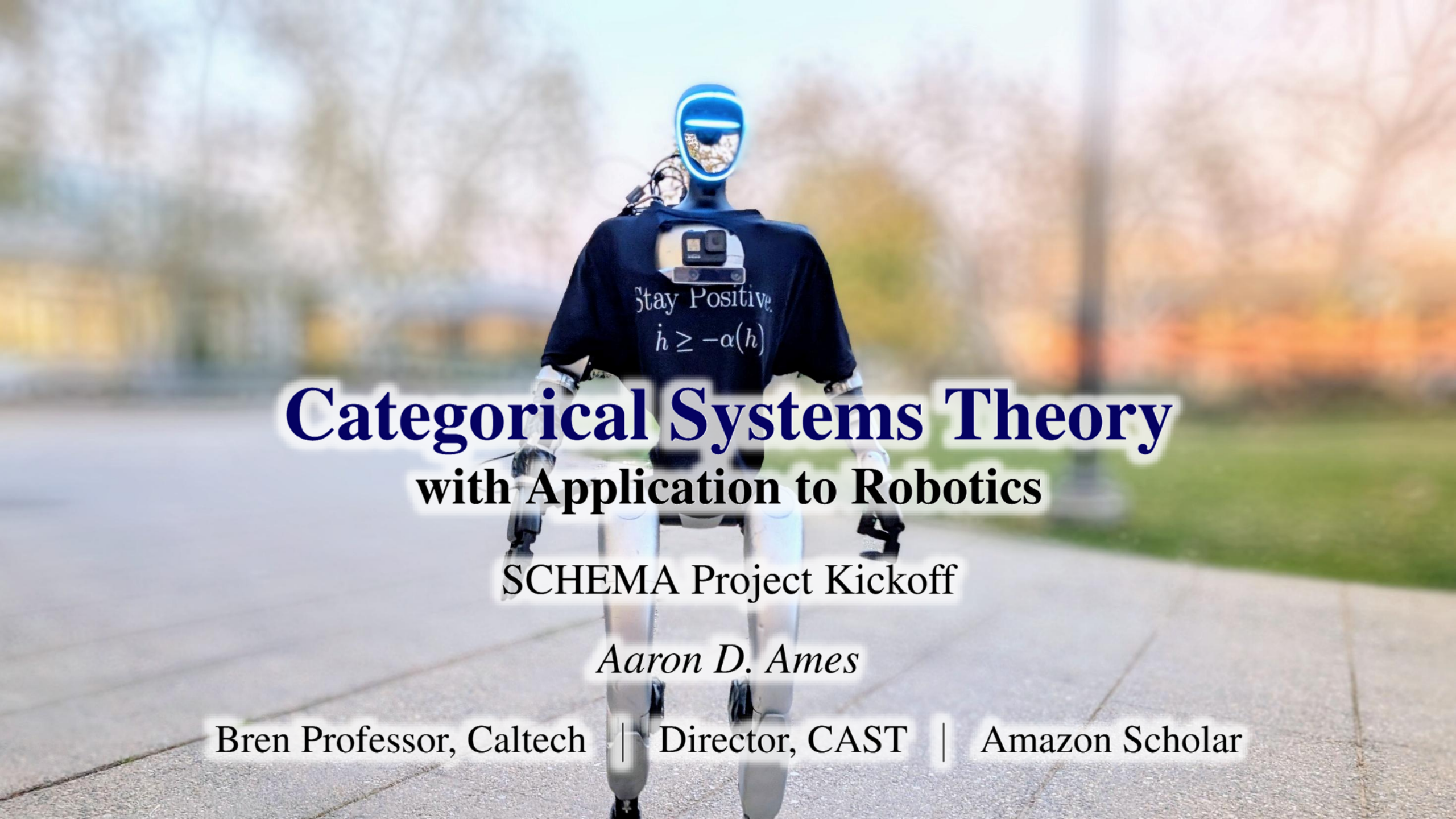


Categorical Systems Theory
with Application to Robotics

SCHEMA Project Kickoff

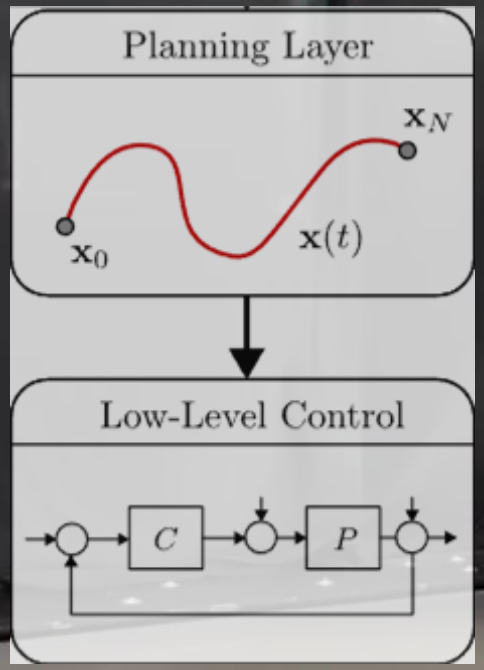
Aaron D. Ames

Bren Professor, Caltech | Director, CAST | Amazon Scholar

Layered Control Architectures

CLF-RL

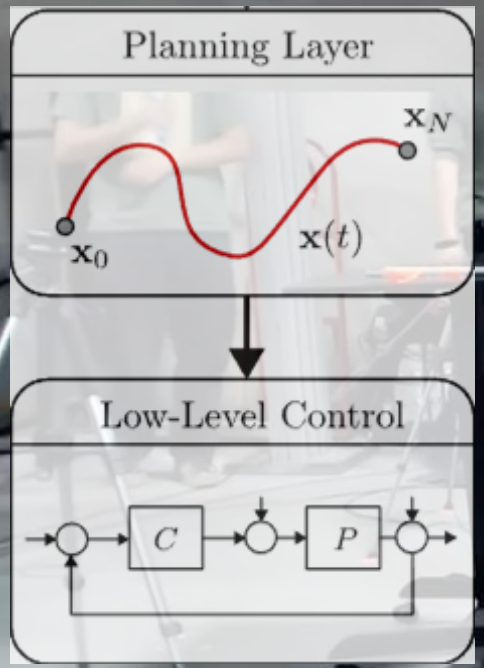
Motor Control



- Li, Olkin, Yue, Ames, RAL 2026
- Olkin, Li, Compton, Ames, IROS 2026

Slow Motion (1/4x)

LIMBO



CBF-RL

Motor Control

0.95 m height

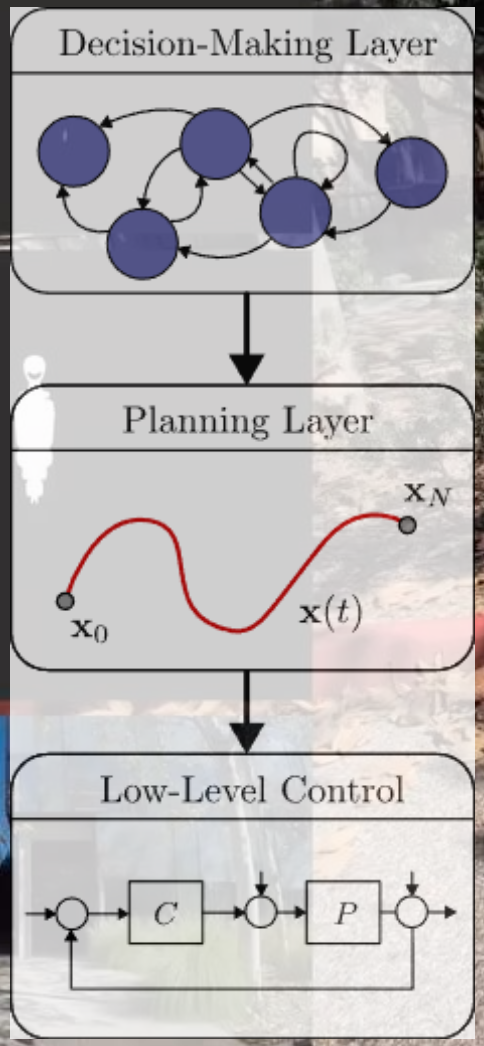
Gonzales, Alvarez, Chen, Ames, Ratliff, Nambi ArXiv 2026

amazon robotics

CBF-MPC

CLF-RL

RL



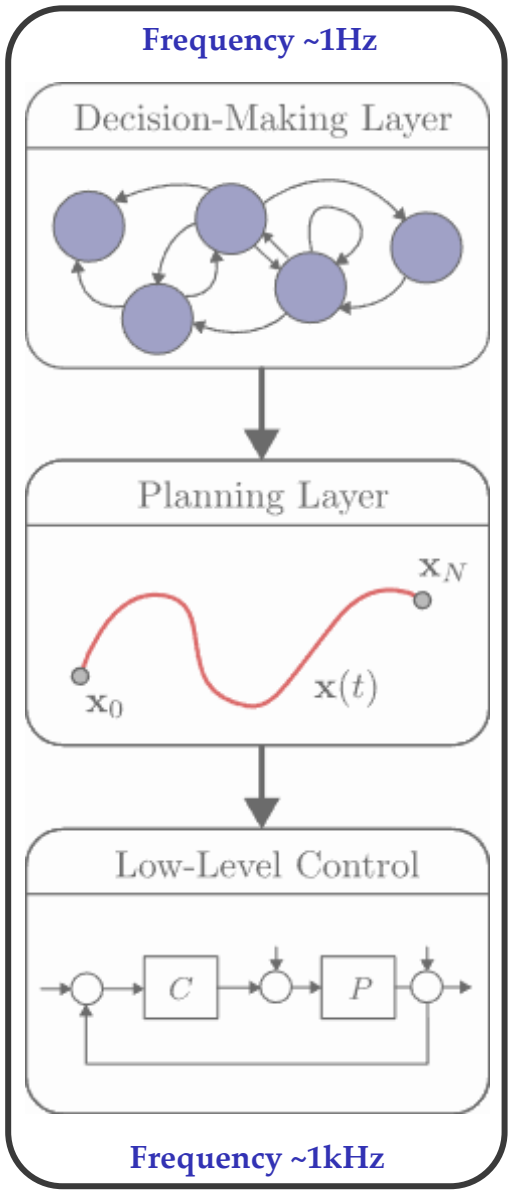
Yamaguchi, Bena, Bahati, Ames, ArXiv 2026

Caltech

LCA: Fractured Landscape

Caltech

Slow



Fast

Formal Methods Worldview:

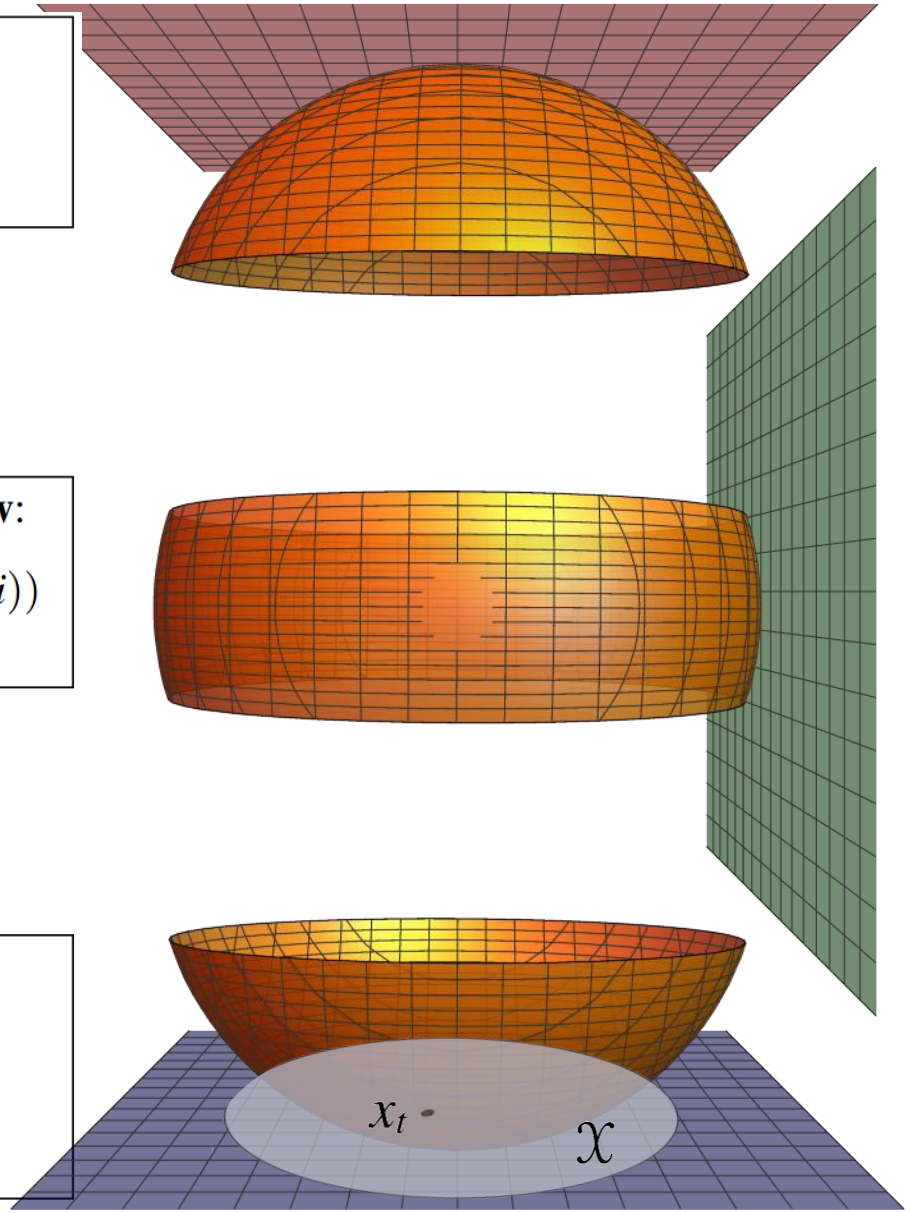
Specifications: $\xi(s) \models \varphi,$
Dynamics: $s' \in f_{\text{FTS}}(s, a)$

Model Predictive Control Worldview:

Dynamics: $z(i+1) = f_{\text{plan}}(z(i), u_{\text{ff}}(i))$
Constraints: $z(k) \in \Pi(\mathcal{X}), \forall k \geq 0$

Nonlinear Control Worldview:

Dynamics: $\dot{x}_t = f(x_t, u_{\text{fb}}(x_t))$
Constraints: $x_t \in \mathcal{X}$
$u_{\text{fb}}(x_t) \in \mathcal{U}, \ \forall t \geq 0$



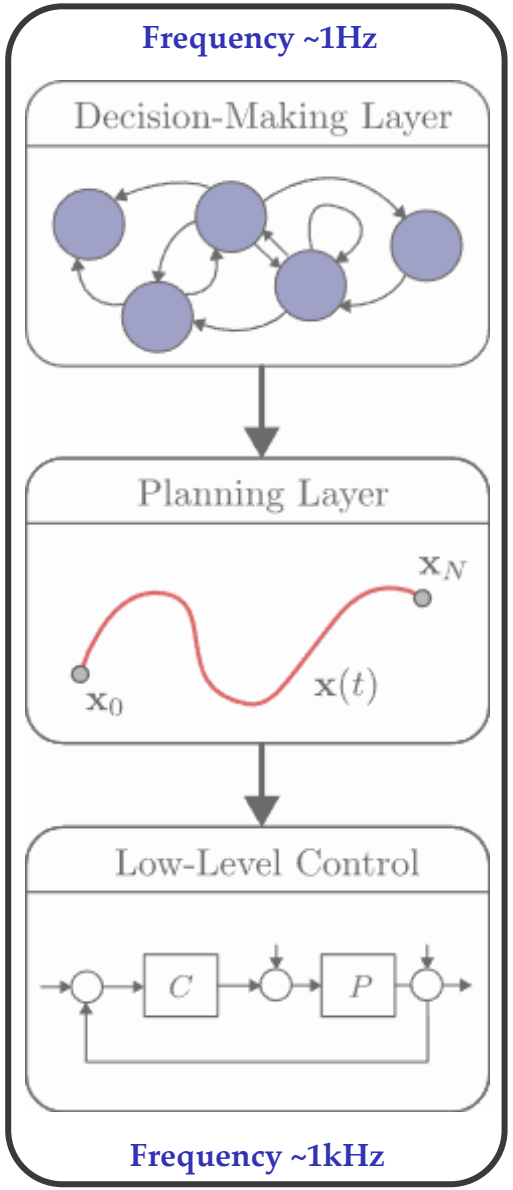
Rosolia, Singletary, AA, TAC 2022

LCA: Need to Unify

Overall LCA Problem

$\xi(x) \models \varphi,$
subject to: $\dot{x}_t = f(x_t, u(x_t))$.

Slow



Fast

Formal Methods Worldview:

Specifications: $\xi(s) \models \varphi,$
Dynamics: $s' \in f_{\text{FTS}}(s, a)$

Top Interface
$z(k) \in s(\lfloor k/\delta \rfloor)$

Model Predictive Control Worldview:

Dynamics: $z(i+1) = f_{\text{plan}}(z(i), u_{\text{ff}}(i))$
Constraints: $z(k) \in \Pi(\mathcal{X}), \forall k \geq 0$

Bottom Interface
$z(k) = \Pi(x_{k\tau})$

Nonlinear Control Worldview:

Dynamics: $\dot{x}_t = f(x_t, u_{\text{fb}}(x_t))$
Constraints: $x_t \in \mathcal{X}$
$u_{\text{fb}}(x_t) \in \mathcal{U}, \ \forall t \geq 0$

Satisfy Specifications

$$\underset{(s,a)}{\text{satisfy}} \quad \xi(s,a) := (L(s(k), a(k)))_{k \in \mathbb{N}} \models \varphi$$
$$\text{s.t.} \quad z(k) \in s(\lfloor k/\delta \rfloor)$$

Interface

Optimize Trajectories

$$\underset{z(k), u_{\text{ff}}(k)}{\text{minimize}} \quad \sum_{i=k}^{k+N-1} \text{Cost}(z(i), u_{\text{ff}}(i)) + \text{Cost}_N(z(k+N))$$
$$\text{s.t.} \quad z(i+1) = f_{\text{DT}}(z(i), u_{\text{ff}}(i))$$
$$z(k) = \Pi(x_{k\tau})$$
$$z(i) \in \Pi(\mathcal{X}) \cap s(\lfloor i/\delta \rfloor)$$

Interface

Optimize Real-Time Feedback

$$\underset{u = u_{\text{ff}} + u_{\text{fb}}}{\text{minimize}} \quad \int_{k\tau}^{(k+N)\tau} \left(\|x(s) - z(\lfloor s/\tau \rfloor)\|_Q^2 + \|u_{\text{fb}}\|_R^2 \right) ds$$
$$\text{s.t.} \quad \dot{x} = f(x, u_{\text{ff}} + u_{\text{fb}})$$
$$x \in \mathcal{X}$$
$$u_{\text{ff}} + u_{\text{fb}} \in \mathcal{U}$$

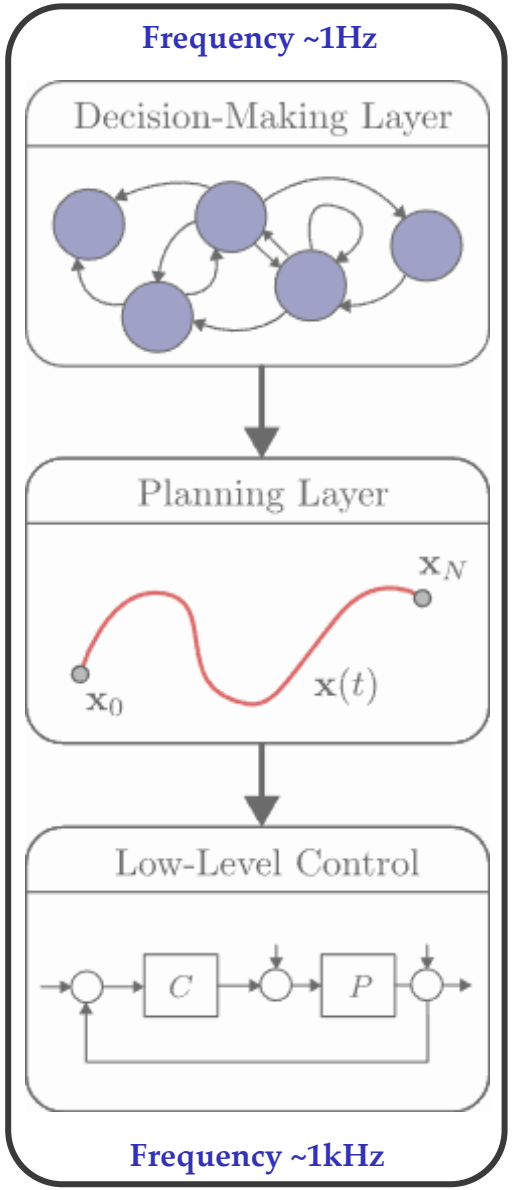
Matni, AA, Doyle, CSM 2024

Idea: Unify via Lyapunov

Caltech

Slow

Fast



Labeled Transition System:

Dynamics: $s' \in f_{\text{FTS}}(s, a)$
Constraints: $s \in \mathcal{S}$

Lyapunov Condition (TS):

$$\max_{s' \in f_{\text{FTS}}(s,a)} V(s') \leq V(s)$$

Discrete-Time Dynamical System:

Dynamics: $z(i+1) = f_{\text{plan}}(z(i))$
Constraints: $z(k) \in \mathcal{Z}$

Lyapunov Condition (DT):

$$\nabla V(z) = V(f_{\text{plan}}(z)) - V(z) \leq 0$$

Continuous-Time Dynamical System:

Dynamics: $\dot{x}(t) = f(x(t))$
Constraints: $x(t) \in \mathcal{X}$

Lyapunov Condition (CT):

$$\dot{V}(x) = \frac{\partial V}{\partial x}\Big|_x f(x(t)) \leq 0$$

Categorical Lyapunov Theory

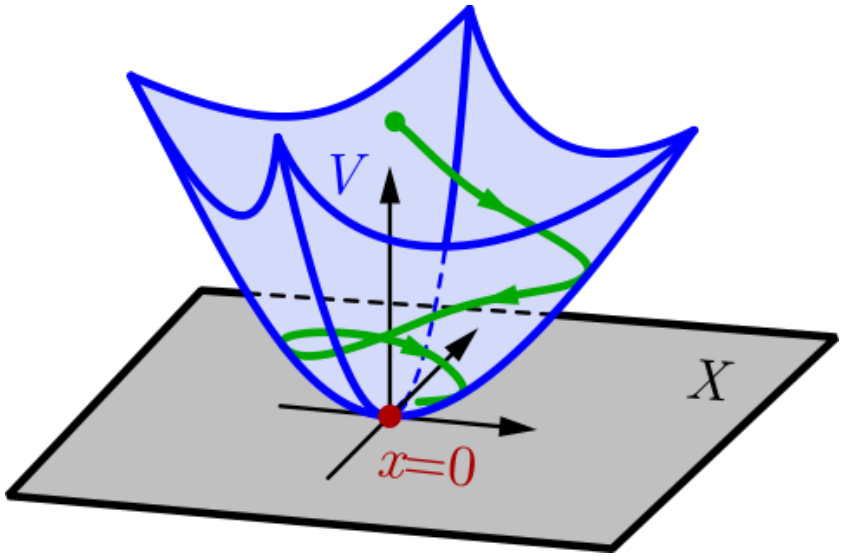
Caltech

Classical Conditions

Characterizing **Solutions** and **Lyapunov** via a diagram:

$$\begin{array}{ccccc} \mathbb{R}\times\mathbb{R} & \xrightarrow{Tc} & \mathcal{X}\times\mathbb{R}^n & \xrightarrow{TV} & \mathbb{R}_{\geq 0}\times\mathbb{R} \\ \uparrow{\scriptstyle 1_{\mathbb{R}}} & \circlearrowleft & \uparrow{\scriptstyle \vec{f}} & \leq & \uparrow{\scriptstyle 0_{\mathbb{R}_{\geq 0}}} \\ \mathbb{R} & \xrightarrow[c]{} & \mathcal{X} & \xrightarrow[V]{} & \mathbb{R}_{\geq 0} \end{array}$$

$\boxed{\dot{c}(t) = f(c(t))}$



unit clock $1_{\mathbb{R}} = (\mathrm{id}_{\mathbb{R}}, 1)$, zero vector field $0_{\mathbb{R}_{\geq 0}}(r) = (r, 0)$.

Theorem

The system $\dot{x} = f(x)$ is stable iff the diagram (lax) commutes.

Lyapunov Condition (CT):

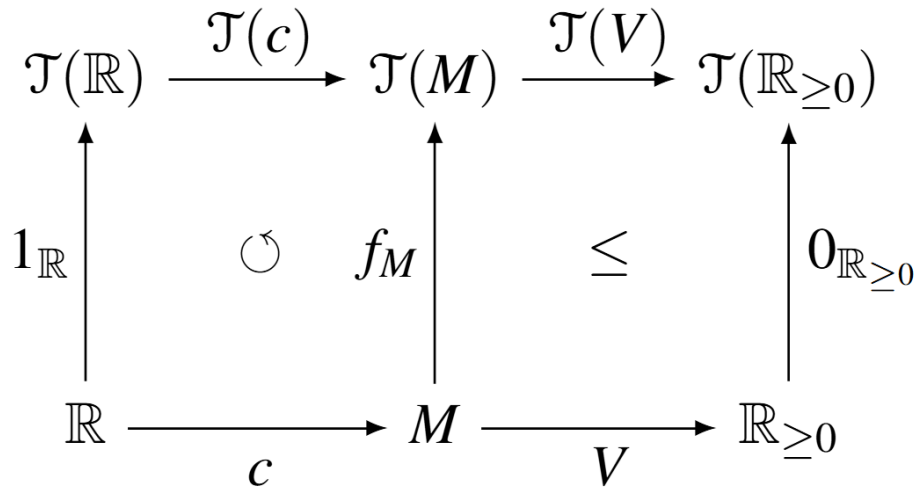
$$\dot{V}(x) = \frac{\partial V}{\partial x}\Big|_x f(x(t)) \leq 0$$

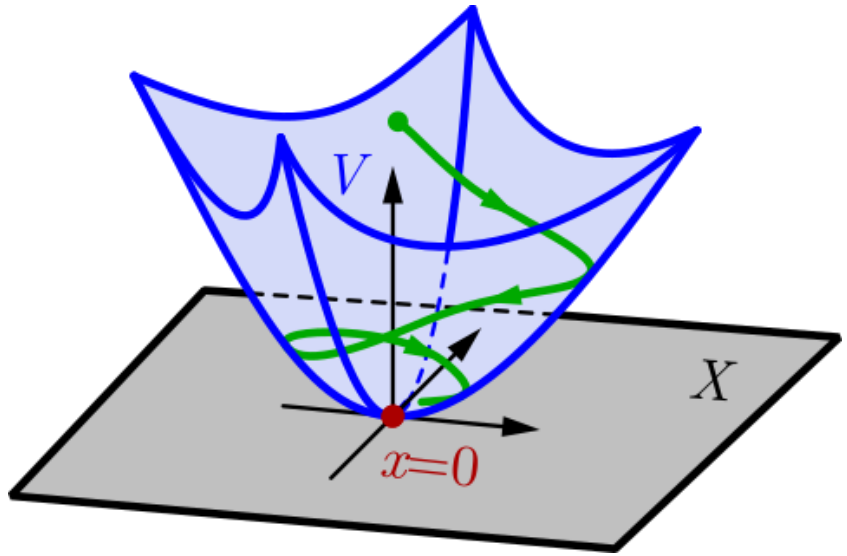
Categorical Lyapunov Theory

Caltech

Vector Fields

Characterizing **Solutions** and **Lyapunov** via a diagram:





unit clock $1_{\mathbb{R}} = (\mathrm{id}_{\mathbb{R}}, 1)$, zero vector field $0_{\mathbb{R}_{\geq 0}}(r) = (r, 0)$.

Theorem

The system $f_M : M \to \mathcal{T}(M)$ is stable iff the diagram (lax) commutes.

Lyapunov Condition (Man):

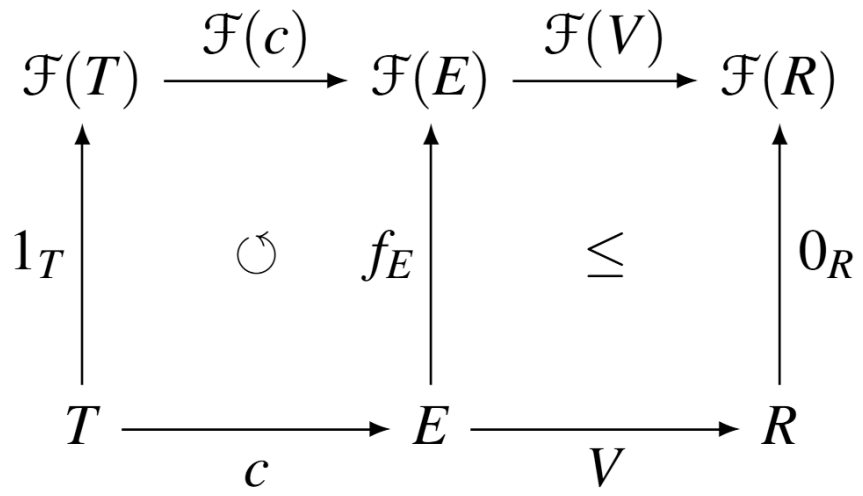
$$\mathcal{T}(V) \circ f_M \leq 0_{\mathbb{R}_{\geq 0}} \circ V$$

$\mathcal{T} : \mathsf{Man} \to \mathsf{Man}$ is the tangent bundle functor, f_M a vector field.

Categorical Lyapunov Theory

Coalgebras

Characterizing **Solutions** and **Lyapunov** via a diagram:



unit clock $1_T : T \to \mathcal{F}(T)$, zero system $0_R : R \to \mathcal{F}(R)$

Theorem

The system $f_E : E \to \mathcal{F}(E)$ is stable iff the diagram (lax) commutes.

$\mathcal{F} : \mathsf{C} \to \mathsf{C}$ an endofunctor, $T =$ time and $R =$ measurement.

On the Stability of Zeno Equilibria*

Aaron D. Ames[1], Paulo Tabuada[2], and Shankar Sastry[1]

HSCC 2006

CATEGORICAL LYAPUNOV THEORY I: STABILITY OF FLOWS

AARON D. AMES, JOE MOELLER, AND PAULO TABUADA

CATEGORICAL LYAPUNOV STABILITY II: STABILITY OF SYSTEMS

AARON D. AMES, SÉBASTIEN MATTENET, AND JOE MOELLER

ArXiv 2025

Lyapunov Condition (C):

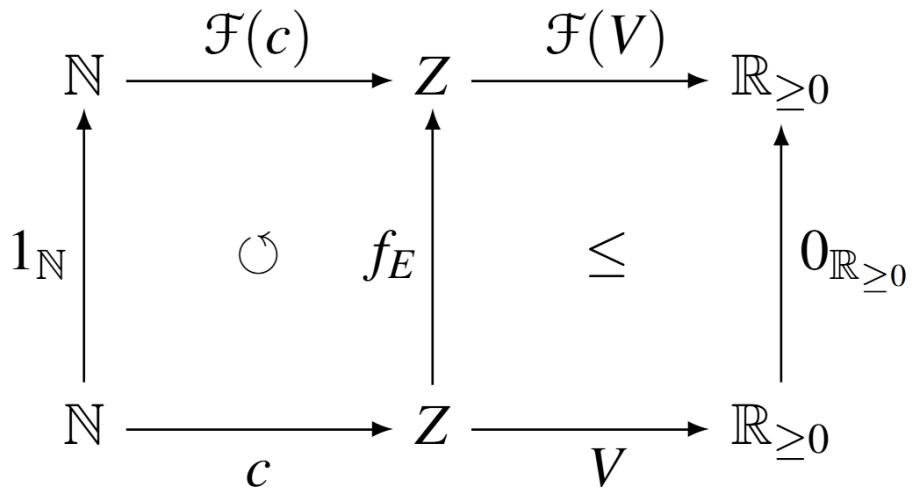
$$\mathcal{F}(V) \circ f_E \leq 0_R \circ V$$

Categorical Lyapunov Theory

Caltech

Discrete Time

Characterizing **Solutions** and **Lyapunov** via a diagram:



unit clock $1_{\mathbb{N}}(k) = k + 1$, zero system $0_{\mathbb{R}_{\geq 0}}(r) = r$.

Theorem

The system $z_{k+1} = f_{\text{plan}}(z_k)$ is stable iff the diagram (lax) commutes.

$\mathcal{F} = I$, $\mathsf{C} = \mathsf{Man}$, $T = \mathbb{N}$ and $R = \mathbb{R}_{\geq 0}$.

Lyapunov Condition (DT):

$$\nabla V(z) = V(f_{\text{plan}}(z)) - V(z) \leq 0$$

Lyapunov Condition (C):

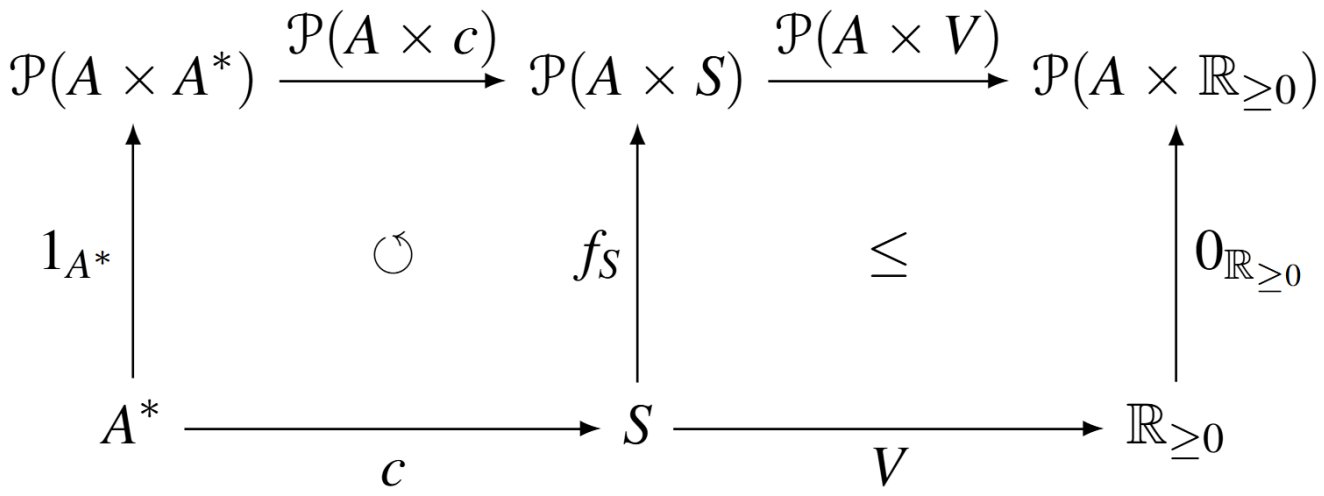
$$\mathcal{F}(V) \circ f_E \leq 0_R \circ V$$

Categorical Lyapunov Theory

Caltech

Transition Systems[1]

Characterizing **Solutions** and **Lyapunov** via a diagram:



unit clock $1_{A^*}(a^*) = \{(a, a^*a)\}_{a \in A}$, zero system $0_{\mathbb{R}_{\geq 0}}(r) = A \times \{r\}$.

Theorem

The system $s' \in f_{\text{FTS}}(s, a)$ is stable iff the diagram (lax) commutes.

$\mathcal{F} = \mathcal{P}(A \times -)$, $\mathsf{C} = \mathsf{Set}$, $T = A^*$ (free monoid) and $R = \mathbb{R}_{\geq 0}$.

Lyapunov Condition (TS):

$$\max_{s' \in f_{\text{FTS}}(s,a)} V(s') \leq V(s)$$

$$f_{\text{FTS}}(s, a) = \{s' : (a, s') \in f_S(s)\}$$

Lyapunov Condition (DT):

$$\nabla V(z) = V(f_{\text{plan}}(z)) - V(z) \leq 0$$

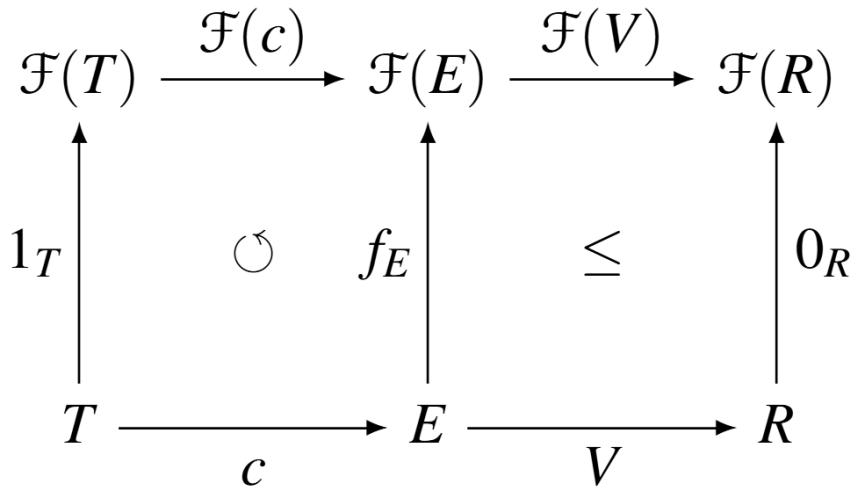
Lyapunov Condition (C):

$$\mathcal{F}(V) \circ f_E \leq 0_R \circ V$$

Categorical Lyapunov Theory ↔ Ability to Generalize

Caltech

$$\text{system } f_E : E \to \mathcal{F}(E)$$



Lyapunov Condition (ℂ):

$$\mathcal{F}(V) \circ f_E \leq 0_R \circ V$$

Lyapunov Condition (TS):

$$\max_{s' \in f_{\text{FTS}}(s,a)} V(s') \leq V(s)$$

$$\text{system } s' \in f_{\text{FTS}}(s,a)$$

Lyapunov Condition (DT):

$$\nabla V(z) = V(f_{\text{plan}}(z)) - V(z) \leq 0$$

$$\text{system } z_{k+1} = f_{\text{plan}}(z_k)$$

Lyapunov Condition (CT):

$$\dot{V}(x) = \frac{\partial V}{\partial x}\Big|_x f(x(t)) \leq 0$$

$$\text{system } \dot{x} = f(x)$$

How do we extend this idea to control systems?

Motivation: Safety Filters

Caltech

Control barrier function

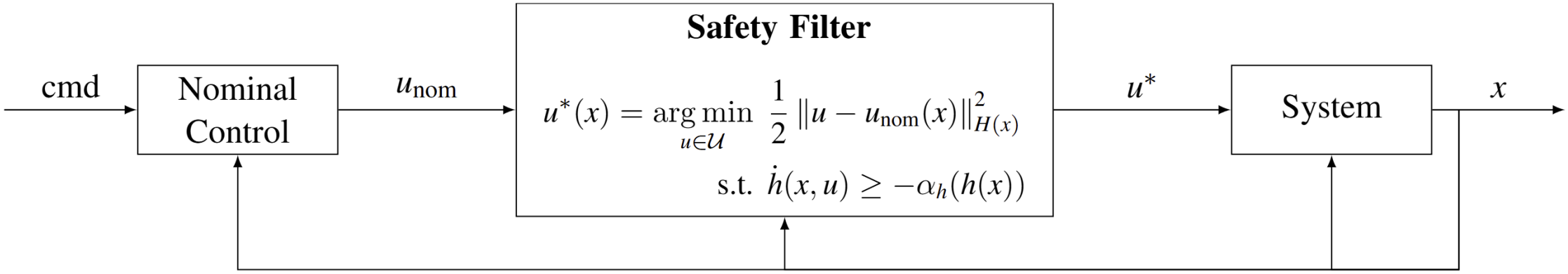
- Safe Set: $S = \{x \in M : h(x) \geq 0\}$,
- CBF condition: $\forall\, x \in M, \exists\, u \in \mathcal{U}$ s.t.

$$\dot{h}(x,u) \geq -\alpha_h(h(x))$$

Theorem (Ames, Grizzle, Tabuada)

CBF $\iff$ *safety (forward invariance of S).*





Motivation: Safety Filters

Caltech

Control barrier function

- Safe Set: $S = \{x \in M : h(x) \geq 0\}$,
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Theorem (Ames, Grizzle, Tabuada)

CBF $\iff$ *safety (forward invariance of S).*

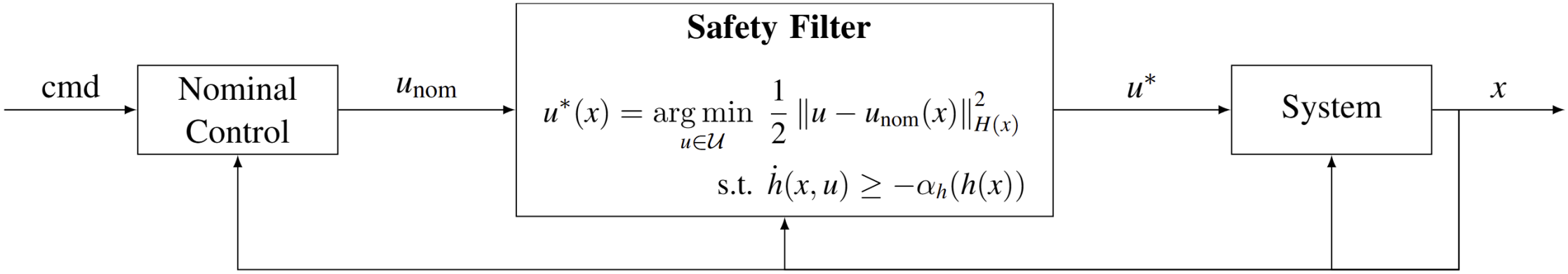
Control Lyapunov function

- Positive definite function: $V(x) > 0\ \forall x \neq 0$.
- CLF condition: $\forall\, x \in M, \exists\, u \in \mathcal{U}$ s.t.

$$\dot{V}(x,u) \leq -\alpha_V(V(x))$$

Theorem (Sontag, Artstein)

CLF $\iff$ *stabilizable* ($\|x(t)\| \leq \beta(t, \|x_0\|)$).

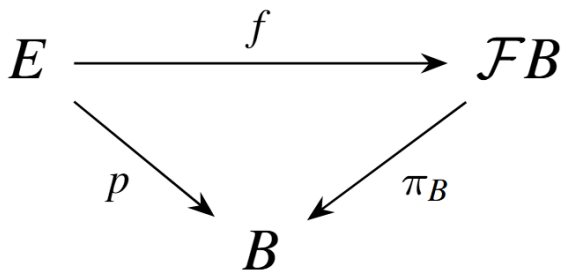


Control Coalgebras

Caltech

Definition

For an endofunctor $\mathcal{F}\colon \mathsf{C} \to \mathsf{C}$, natural transformation $\pi\colon \mathcal{F} \Rightarrow \mathrm{id}_{\mathsf{C}}$, a **control coalgebra**:

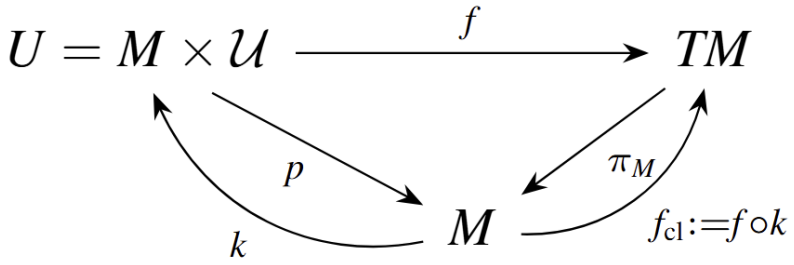


denoted by $(p\colon E \to B, f)$, consists of

1. **state space** $B \in \mathsf{C}$
2. **control bundle** $p\colon E \to B$
3. **control system** $f\colon E \to \mathcal{F}B$

Continuous time

Tangent bundle functor $T : \mathsf{Man} \to \mathsf{Man}$ yields:



If $U = M \times \mathcal{U}$, obtain classical control system:

$$\dot{x} = f(x,u), \qquad u \in \mathcal{U}.$$

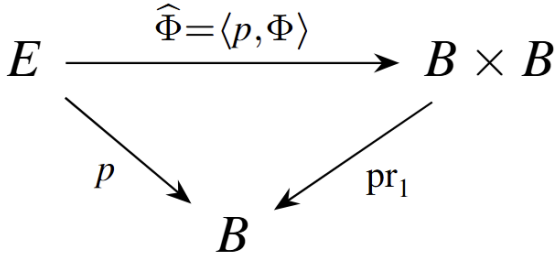
- feedback controller is a smooth section: $k(x) = (x, \kappa(x))$,
- closed loop system: $f_{\mathrm{cl}}(x) = f(x, \kappa(x))$.

Control Coalgebras

Caltech

Discrete time

$\mathcal{F}_{\mathrm{d}}(X) := X \times X$ yielding control coalgebra:



$$\widehat{\Phi}(x,u) = \big(x, \Phi(x,u)\big)$$

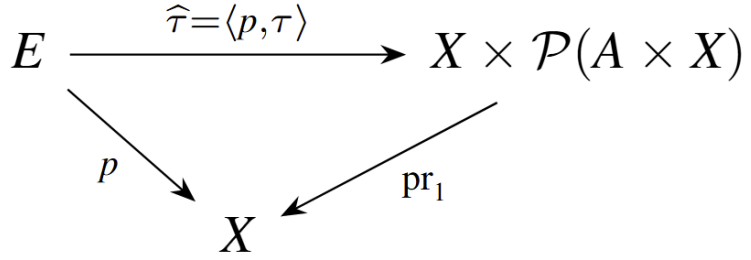
If $E = B \times \mathcal{U}$, obtain discrete-time system:

$$x_{k+1} = \Phi(x_k, u_k).$$

- feedback controller: $k(x) = (x, \kappa(x))$,
- closed loop system: $x_{k+1} = \Phi(x_k, \kappa(x_k))$.

Controlled LTS

$\mathcal{F}_A(X) := X \times \mathcal{P}(A \times X)$ yielding control coalgebra:



$$\widehat{\tau}(e) = \big(p(e), \tau(e)\big)$$

Fiber $E_x = p^{-1}(x)$, control choices at x:

$$(a_k, x_{k+1}) \in \tau(e_k) \qquad e_k \in E_{x_k}$$

- feedback controller: $k : X \to E$,
- closed loop system: $\widehat{\tau}_{\mathrm{cl}}(x) = \big(x, \tau(k(x))\big)$.

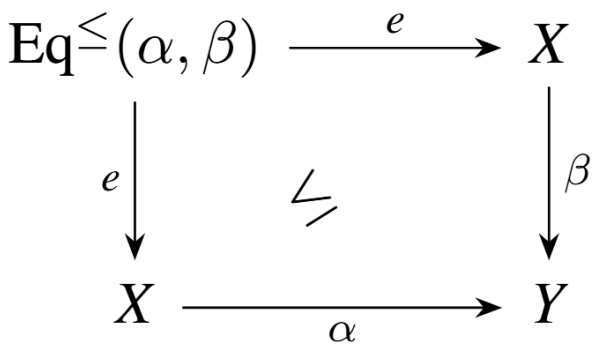
Anchored endofunctors: take $\mathcal{G} : \mathsf{C} \to \mathsf{C}$ to $\widehat{\mathcal{G}}(X) := X \times \mathcal{G}X$.

Lax Equalizers

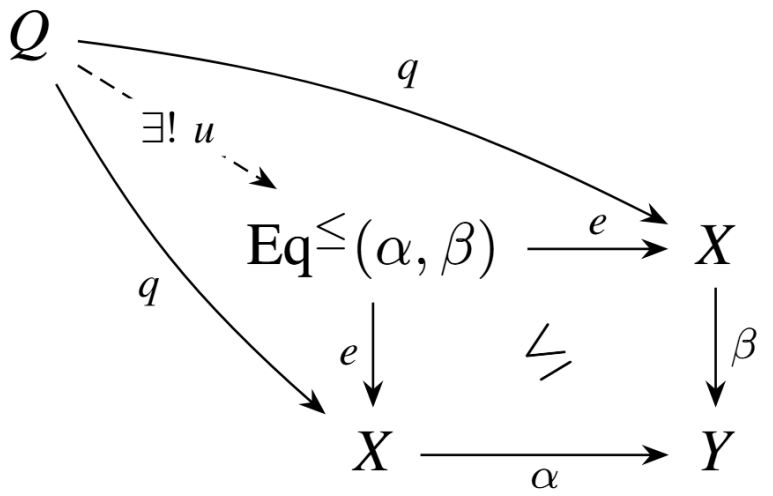
Caltech

Lax equalizer

If Y is posetal, lax equalizer defined by:

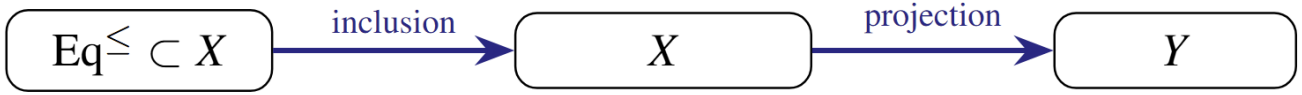


Universal Property



$$x \in \mathrm{Eq}^{\leq}(\alpha,\beta) \iff \alpha(x) \leq \beta(x).$$

$$\alpha \circ q(y) \leq \beta \circ q(y) \implies u(y) \in \mathrm{Eq}^{\leq}(\alpha,\beta)$$

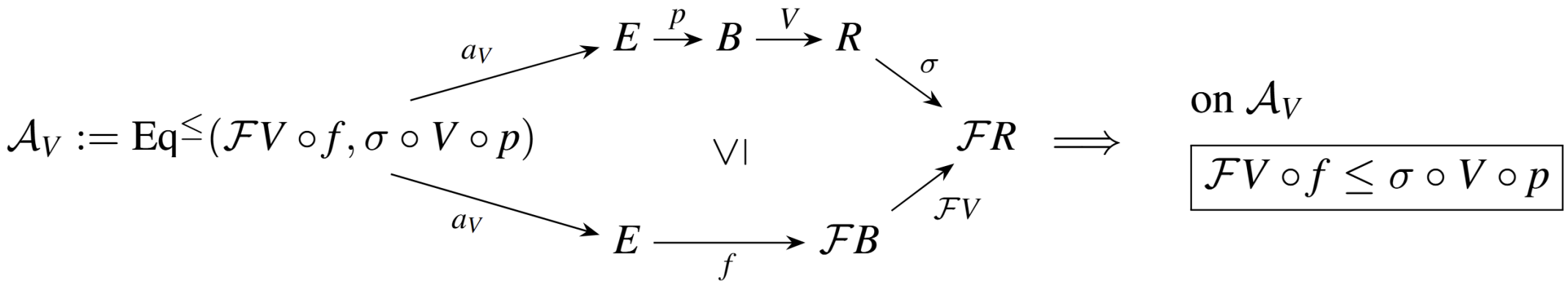


function inequality = lax equalizer subobject

Control Lyapunov Morphisms

Caltech

For a control coalgebra $(p : E \to B, f)$, the **admissible locus** of $V : B \to R$:



Control Lyapunov morphism (CLM)

Positive definite $V : B \to R$ such that:

$$\boxed{\pi_V := p \circ a_V : \mathcal{A}_V \xrightarrow{a_V} E \xrightarrow{p} B}$$

is a **point cover**: π_V an *epi*, $\mathsf{C}(1, \pi_V)$ surjective.

Implications

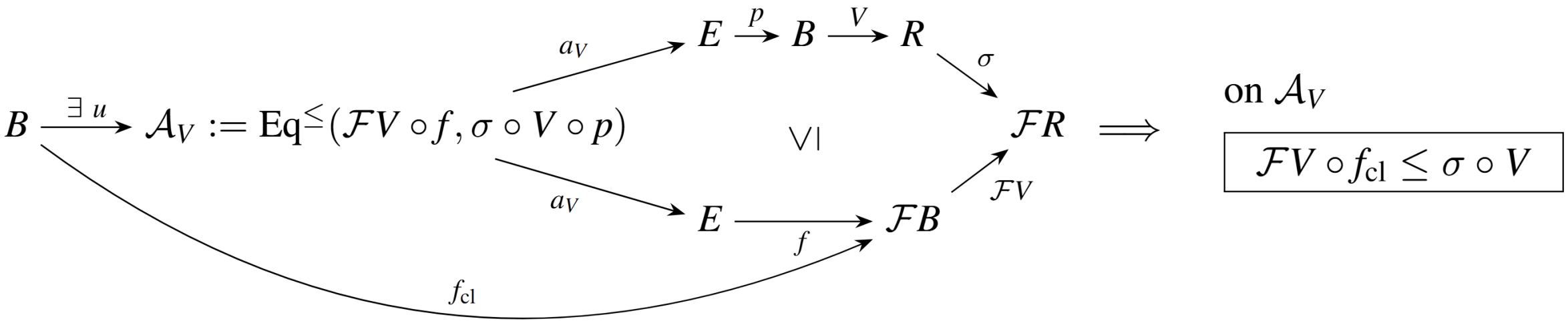
Recover the key CLF idea: for all $x \in B$

$$\exists e_x \in E_x \qquad \text{s.t.} \qquad \underbrace{\mathcal{F}V \circ f(e_x)}_{\text{“}\dot{V}(x,u)\text{”}} \leq \underbrace{\sigma \circ V(x)}_{\text{“}-\alpha(V(x))\text{”}} .$$

Control Lyapunov Theorem

Caltech

For a control coalgebra $(p : E \to B, f)$, if $V : B \to R$ a control Lyapunov morphism:



Theorem

If $\exists$ a section $u : B \to \mathcal{A}_V$, then $V : B \to R$ is a Lyapunov morphism for the closed loop system:

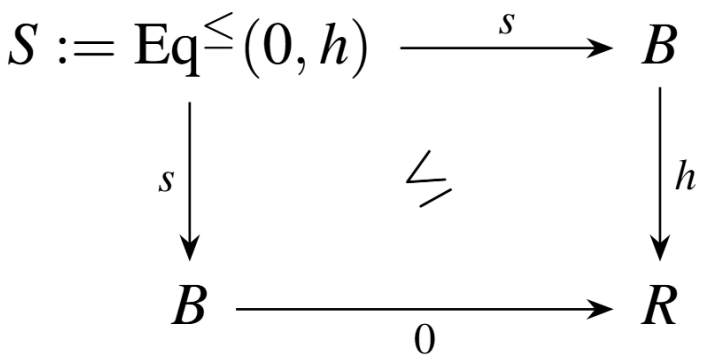
$$f_{\mathrm{cl}} := f \circ a_V \circ u : B \to \mathcal{F}B \qquad \Longrightarrow \qquad \mathcal{F}V \circ f_{\mathrm{cl}} \leq \sigma \circ V$$

CLM + section $\Longrightarrow$ Lyapunov morphism $\Longrightarrow$ Stability

Safety

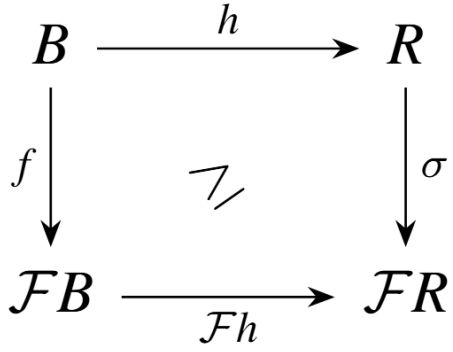
Caltech

Safe subobject



$$\boxed{x \in S \iff h(x) \geq 0}$$

Barrier morphism



$$\boxed{\mathcal{F}h \circ f \geq \sigma \circ h}$$

Theorem (Categorical Nagumo)

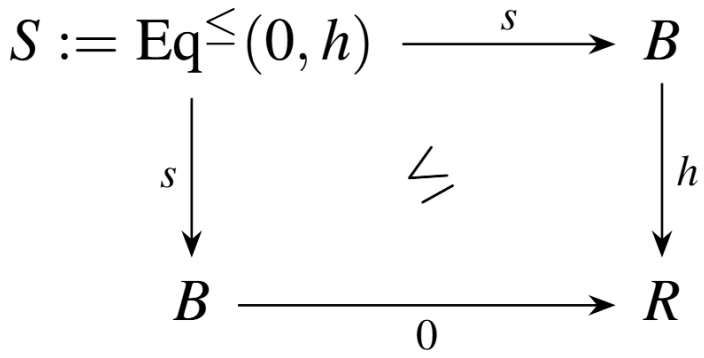
*The subobject $s : S \hookrightarrow B$ is **forward invariant** for $f : B \to \mathcal{F}(B)$ if h is a barrier morphism.*

Recover: $S = \{x \in B : h(x) \geq 0\}$ is safe (forward invariant) for $\dot{x} = f(x)$ if $\boxed{\dot{h} \geq -\alpha(h)}$.

Safety

Caltech

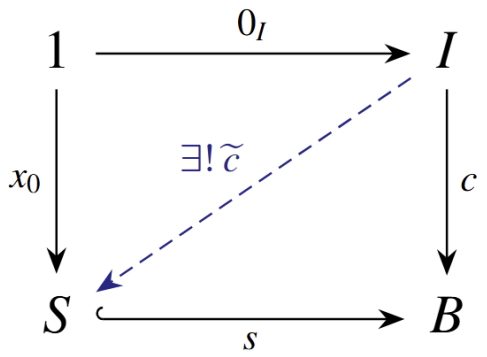
Safe subobject



$$\boxed{x \in S \iff h(x) \geq 0}$$

Forward Invariant

For solutions $c : I \to B$ with $c \circ 0_I = s \circ x_0$:



$$\boxed{c(0) \in S \implies c(t) \in S \quad \forall t \geq 0}$$

Theorem (Categorical Nagumo)

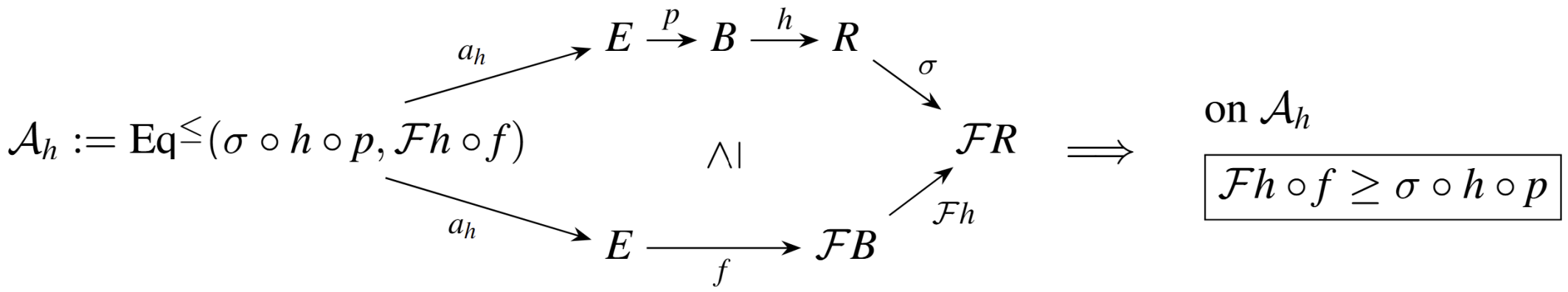
*The subobject $s : S \hookrightarrow B$ is **forward invariant** for $f : B \to \mathcal{F}(B)$ if h is a barrier morphism.*

Recover: $S = \{x \in B : h(x) \geq 0\}$ is safe (forward invariant) for $\dot{x} = f(x)$ if $\boxed{\dot{h} \geq -\alpha(h)}$.

Control Barrier Morphisms

Caltech

For a control coalgebra $(p : E \to B, f)$, the **admissible locus** of $h : B \to R$:



Control barrier morphism (CBM)

$h\colon B \to R$ such that the composite:

$$\pi_h := p \circ a_h : \mathcal{A}_h \xrightarrow{a_h} E \xrightarrow{p} B$$

is a **point cover**: π_h an *epi*, $\mathsf{C}(1, \pi_h)$ surjective.

Implications

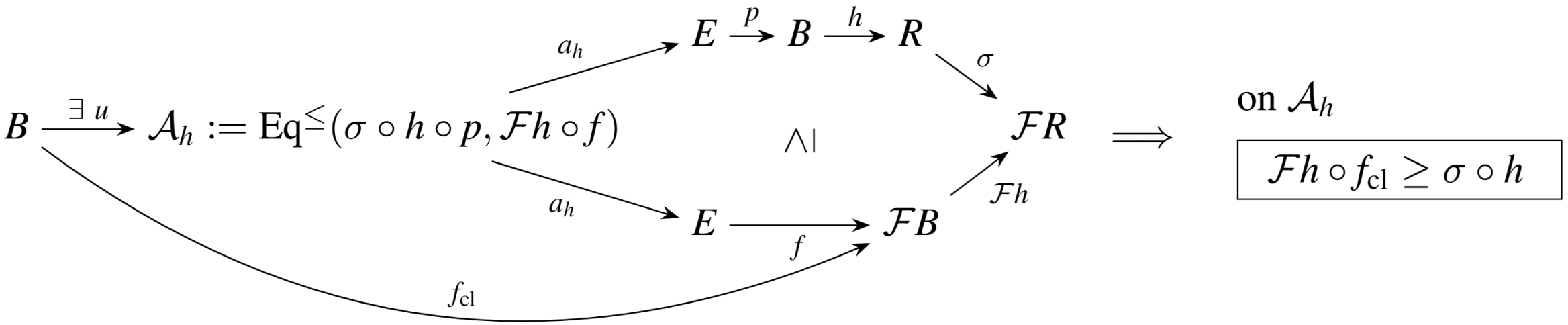
Recover the key CBF idea: for all $x \in B$

$$\exists e_x \in E_x \quad \text{s.t.} \quad \underbrace{\mathcal{F}h \circ f(e_x)}_{\text{“}\dot{h}(x,u)\text{”}} \geq \underbrace{\sigma \circ h(x)}_{\text{“}-\alpha(h(x))\text{”}} .$$

Control Barrier Theorem

Caltech

For a control coalgebra $(p : E \to B, f)$, if $h : B \to R$ a control barrier morphism:



Theorem

If $\exists$ a section $u : B \to \mathcal{A}_h$, then $h : B \to R$ is a barrier morphism for the closed loop system:

$$f_{\mathrm{cl}} := f \circ a_h \circ u : B \to \mathcal{F}B \qquad \Longrightarrow \qquad \mathcal{F}h \circ f_{\mathrm{cl}} \geq \sigma \circ h$$

CBM + section $\Longrightarrow$ barrier morphism $\Longrightarrow$ safety

Safety for Layered Architectures

Caltech

Reduced-order model (RoM)

$$\dot{z} = \nu, \qquad h(z(t)) \geq 0$$

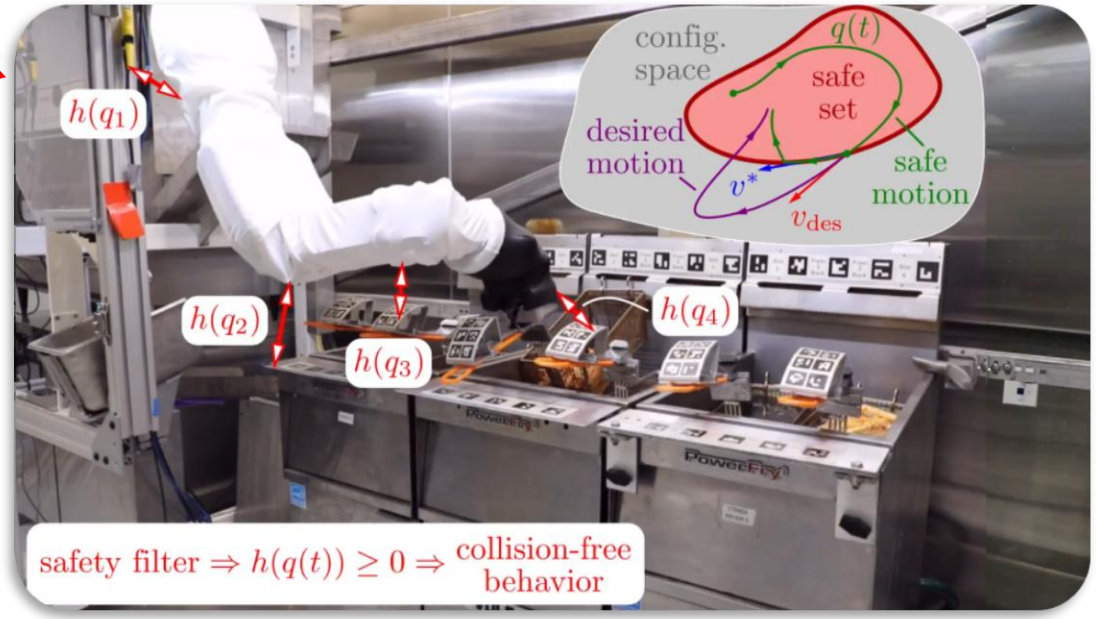
Top interface

z

Safety filter

$$\kappa_{\text{safe}}(z) = \underset{\nu \in \mathbb{R}^n}{\operatorname{argmin}} \frac{1}{2}\|\nu - \nu_{\text{des}}(z)\|^2$$

$$\text{s.t.} \quad Dh(z)\nu \geq -\alpha h(z)$$



Theorem

If $\exists$ *a section* $u : B \to \mathcal{A}_h$, *then* $h : B \to R$ *is a barrier morphism for the closed loop system:*

$$f_{\text{cl}} := f \circ a_h \circ u : B \to \mathcal{F}B \qquad \Longrightarrow \qquad \mathcal{F}h \circ f_{\text{cl}} \geq \sigma \circ h$$

Singletary, Guffey, Molnar, Sinnet, AA, IROS/RAL 2022

Safety for Layered Architectures

Safe set for RoM: $S = \{z \in Z \mid h(z) \geq 0\}$

Safe RoM reference: $\dot{h}(z, \kappa_{\text{safe}}(z)) \geq -\alpha h(z)$

Reduced-order model (RoM)

$$\dot{z} = \nu, \qquad h(z(t)) \geq 0$$

Top interface

z

Safety filter

$$\kappa_{\text{safe}}(z) = \underset{\nu \in \mathbb{R}^n}{\text{argmin}} \frac{1}{2}\|\nu - \nu_{\text{des}}(z)\|^2$$

$$\text{s.t.} \quad Dh(z)\nu \geq -\alpha h(z)$$

Bottom interface

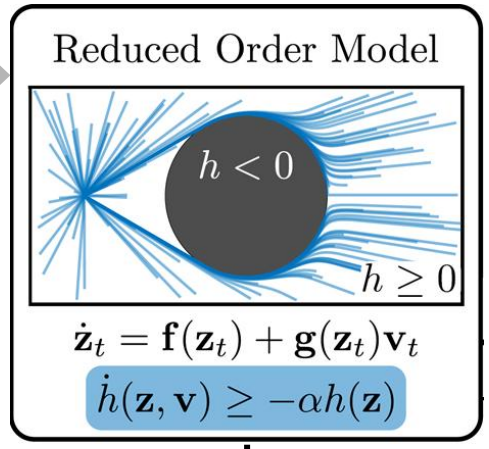
$v = \kappa_{\text{safe}}(z)$

Full-order model (FoM)

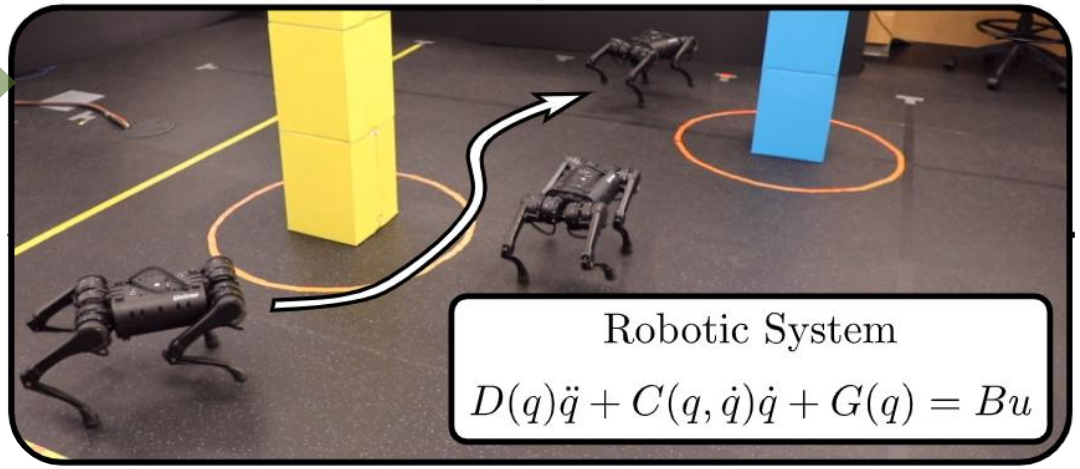
$$D(q)\ddot{q} + C(q, \dot{q}) = Bu$$

$$u = K(q, \dot{q}, v), \quad h(q(t)) \geq 0$$

$z = q$



$\kappa_{\text{safe}}(z)$



RoM Safety $\Longrightarrow$ FoM Safety?

- Molnar, Cosner, Singletary, Ubellacker, AA, ICRA/RAL 2022
- Compton, Cohen, AA, L4DC 2025

Multi-Layer Safety

Reduced-order model (RoM)

$$\dot{z} = \nu, \qquad h(z(t)) \geq 0$$

Top interface

z

Safety filter

$$\kappa_{\text{safe}}(z) = \underset{\nu \in \mathbb{R}^n}{\operatorname{argmin}} \frac{1}{2}\|\nu - \nu_{\text{des}}(z)\|^2$$

$$\text{s.t.} \quad Dh(z)\nu \geq -\alpha h(z)$$

Bottom interface

$v = \kappa_{\text{safe}}(z)$

Full-order model (FoM)

$$D(q)\ddot{q} + C(q,\dot{q}) = Bu$$

$$u = K(q,\dot{q},v), \quad h(q(t)) \geq 0$$

$z = q$

Safe set for RoM: $S = \{z \in Z \mid h(z) \geq 0\}$

Safe RoM reference: $\dot{h}(z, \kappa_{\text{safe}}(z)) \geq -\alpha h(z)$

$z = q, \quad Z = Q$

Tracking error: $e = \dot{q} - \kappa_{\text{safe}}(q)$

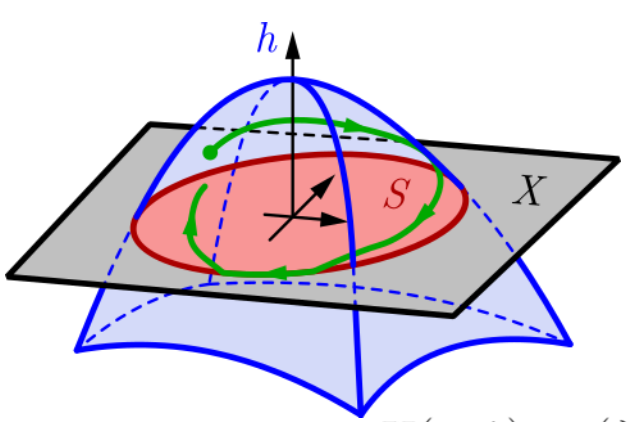
Tracking controller: $u = K(q, \dot{q}, \kappa_{\text{safe}}(q))$

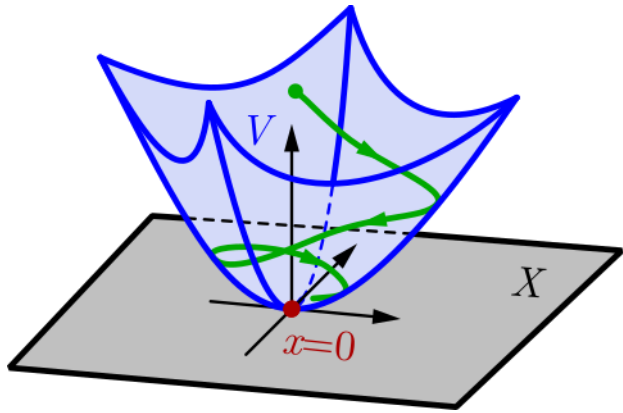
Fast enough tracking: $\dot{V}(q,\dot{q},e) \leq -\lambda V(q,\dot{q})$

Theorem

If $\lambda > \alpha$ *then* $\mathcal{C} = \{(q,\dot{q}) : h(q) \geq 0\}$ *is safe:*

$$H(q_0,\dot{q}_0) \geq 0 \quad \Rightarrow \quad h(q(t)) \geq 0 \quad \forall t \geq 0.$$



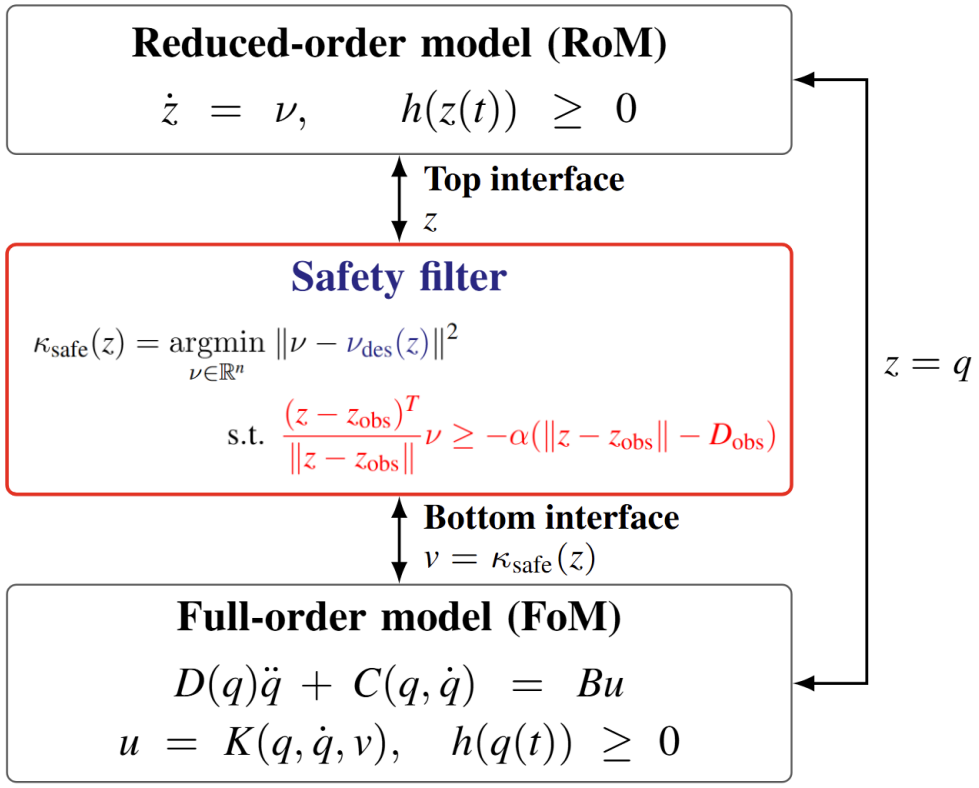


$$H(q,\dot{q}) = (\lambda - \alpha)h(q) - \gamma V(q,\dot{q})$$

- Molnar, Cosner, Singletary, Ubellacker, AA, ICRA/RAL 2022
- Compton, Cohen, AA, L4DC 2025

Safety for Layered Architectures

Caltech



Reduced order Model (RoM): $z \in \mathbb{R}^n$ position, $v \in \mathbb{R}^n$ velocity:

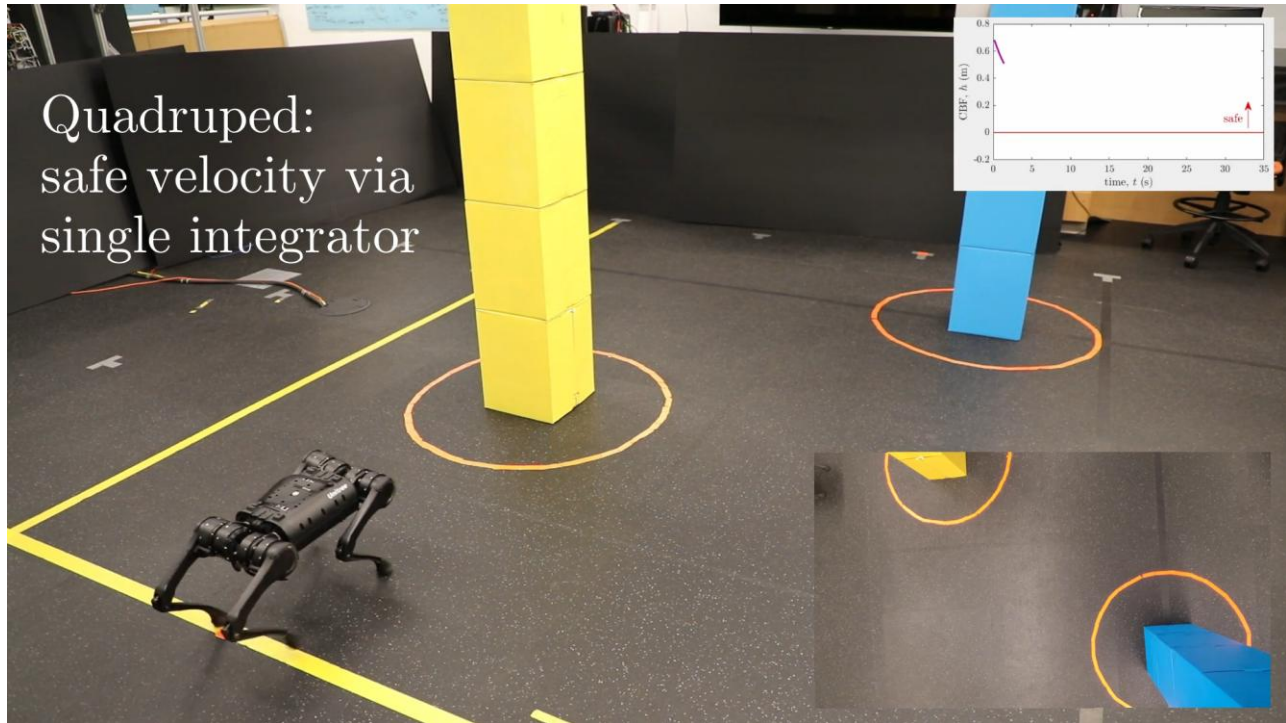
$$\dot{z} = \nu,$$

Barrier function:

$$h(z) = \|z - z_{\text{obs}}\| - D_{\text{obs}} \quad \Rightarrow \quad S = \{z \in \mathbb{R}^n : h(z) \geq 0\}$$

Control barrier function

$$\dot{h}(z, v) = \frac{\partial h}{\partial z}\dot{z} = \underbrace{\frac{(z - z_{\text{obs}})^T}{\|z - z_{\text{obs}}\|}}_{Dh(z)} \underbrace{\nu}_{\dot{z}} \quad \Rightarrow \quad \begin{array}{c} Dh(z)\nu \geq -\alpha(h(z)) \\ \Downarrow \\ S \text{ safe} \end{array}$$



Safety for Layered Architectures

Reduced-order model (RoM)

$$\dot{z} = \nu, \qquad h(z(t)) \geq 0$$

Top interface

z

Safety filter

$$\kappa_{\text{safe}}(z) = \underset{\nu \in \mathbb{R}^n}{\operatorname{argmin}} \|\nu - \nu_{\text{des}}(z)\|^2$$

$$\text{s.t.} \quad \frac{(z - z_{\text{obs}})^T}{\|z - z_{\text{obs}}\|} \nu \geq -\alpha(\|z - z_{\text{obs}}\| - D_{\text{obs}})$$

Bottom interface

$v = \kappa_{\text{safe}}(z)$

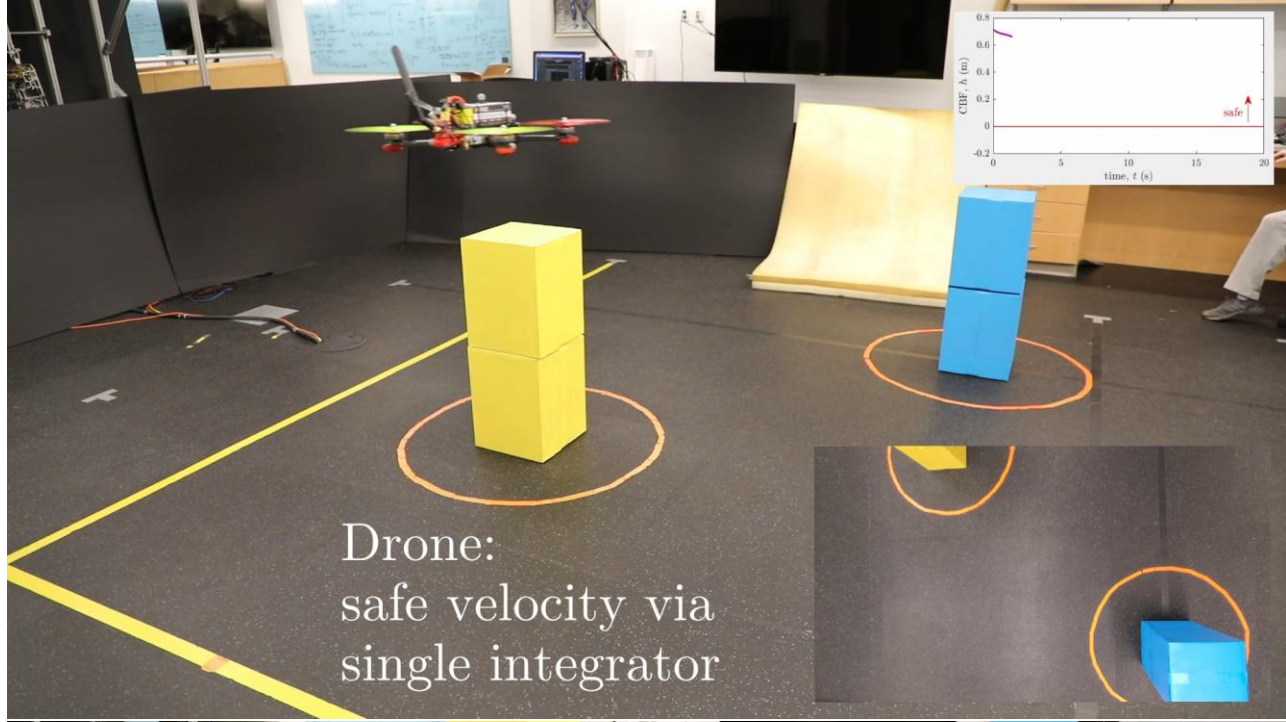
Full-order model (FoM)

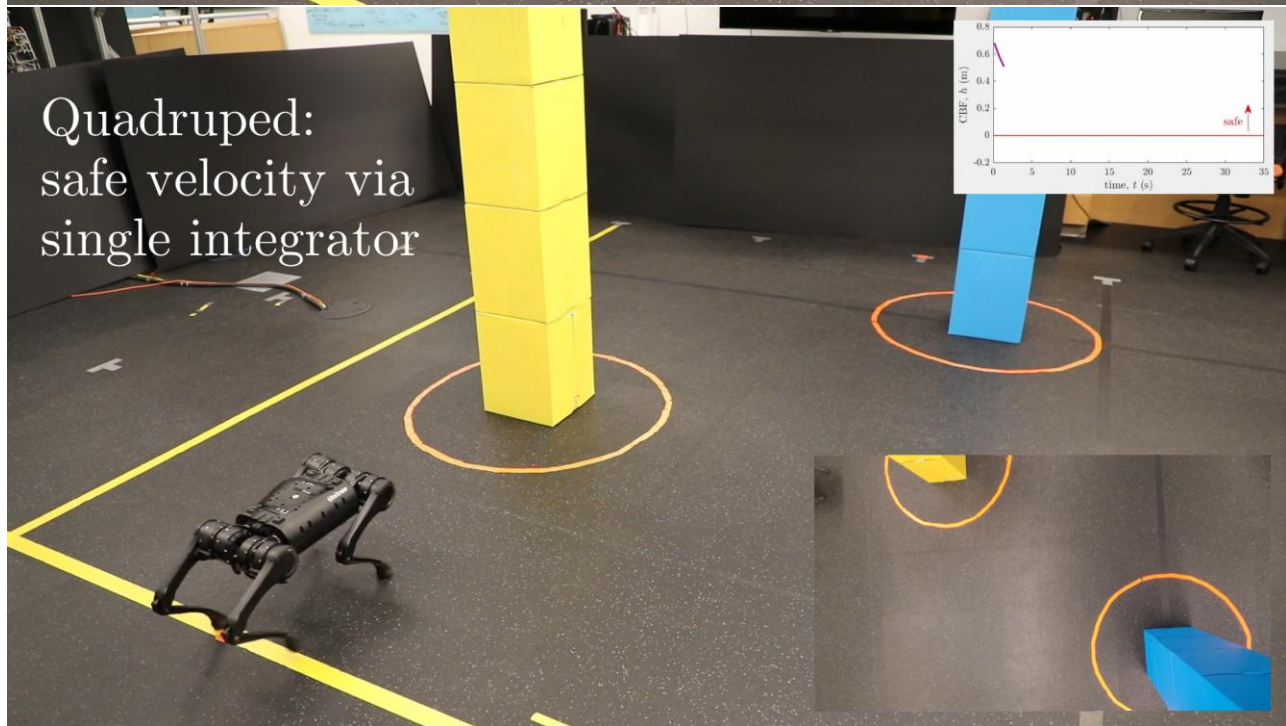
$$D(q)\ddot{q} + C(q, \dot{q}) = Bu$$

$$u = K(q, \dot{q}, v), \quad h(q(t)) \geq 0$$

$z = q$

How do we generalize this idea?





Composition: 20 Years in the Making

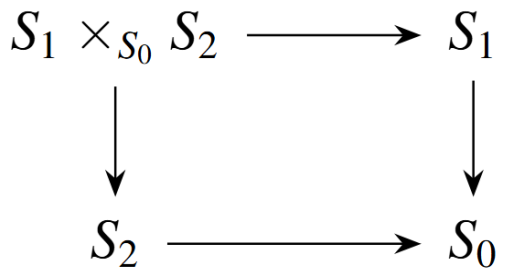
Caltech

2005 workshop / 2006 proceedings

Semantics of embedded systems

Homogeneous Semantics Preserving Deployments of Heterogeneous Networks of Embedded Systems

Aaron D. Ames, Alberto Sangiovanni-Vincentelli and Shankar Sastry



Composition as a limit of tagged systems

Now: systems, inputs, certificates

Layered control architectures

Categorical Lyapunov theory

⇓

Categorical control theory

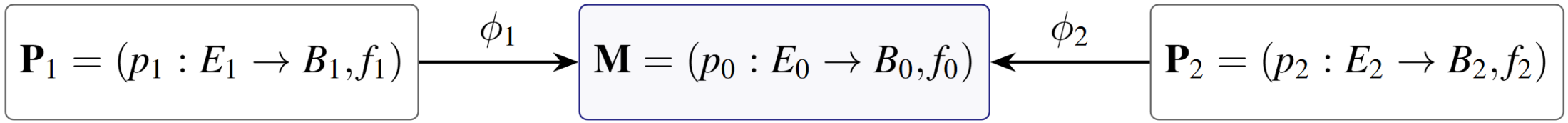
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Layered Control Architectures

Same compositional question;
now for layered control architectures.

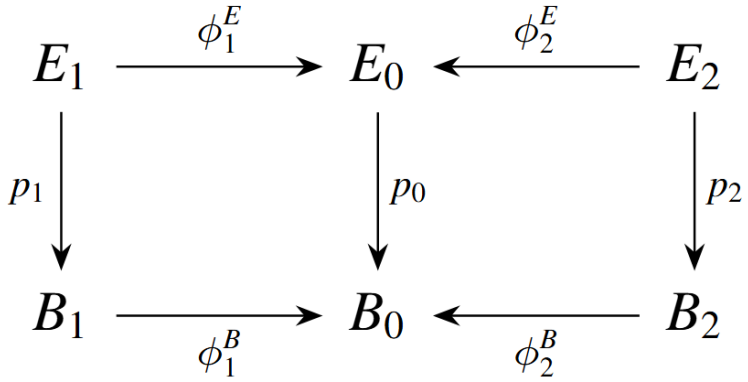
Composition over a Mediator

Caltech

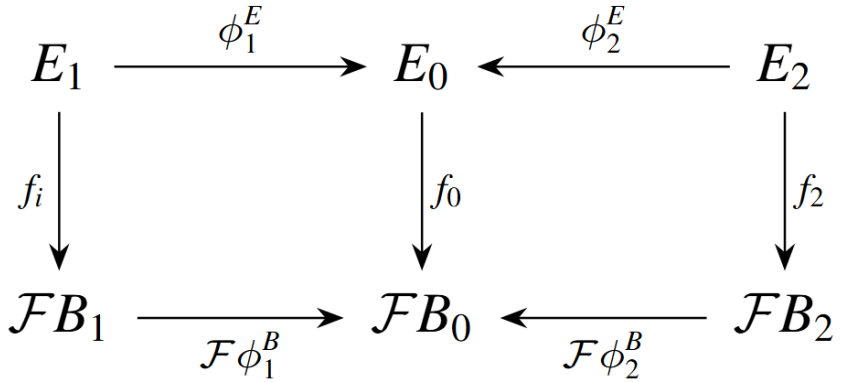


Shared state and port information
matched by the interface maps: $\phi_i = (\phi_i^B, \phi_i^E)$

Strict interface morphisms



Dynamics respect the interface



The architecture is a **cospan** in $\mathsf{Ctrl}_{\mathcal{F}}(\mathcal{C})$.

Composition as the Pullback

Caltech

Given $P_1 \to M \leftarrow P_2$, assume the component pullbacks exists, and $\mathcal{F}$ preserves the state pullback.

State–input space

$$\begin{array}{ccc} E & \xrightarrow{p_1^E} & E_1 \\ {\scriptstyle p_2^E}\downarrow & \lrcorner & \downarrow{\scriptstyle \phi_1^E} \\ E_2 & \xrightarrow[\phi_2^E]{} & E_0 \end{array}$$

State space

$$\begin{array}{ccc} B & \xrightarrow{p_1^B} & B_1 \\ {\scriptstyle p_2^B}\downarrow & \lrcorner & \downarrow{\scriptstyle \phi_1^B} \\ B_2 & \xrightarrow[\phi_2^B]{} & B_0 \end{array}$$

Induced (unique) dynamics

$$\begin{array}{ccc} E & \xrightarrow{p_i^E} & E_i \\ {\scriptstyle \exists!\, f}\downarrow & & \downarrow{\scriptstyle f_i} \\ \mathcal{F}B & \xrightarrow[\mathcal{F}p_i^B]{} & \mathcal{F}B_i \end{array}$$

Proposition

Composition is the pullback:

$$\boxed{\mathbf{P} := \mathbf{P}_1 \times_{\mathbf{M}} \mathbf{P}_2.}$$

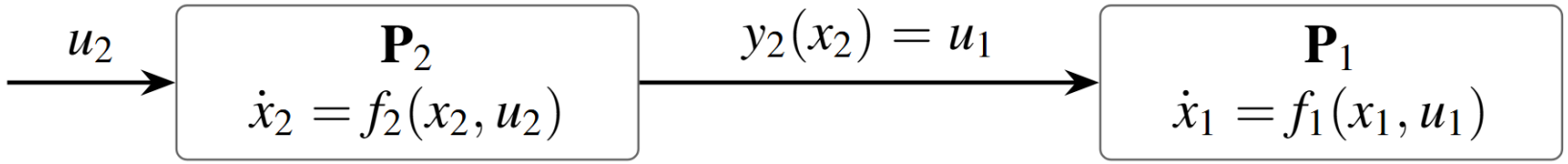
where $\mathbf{P} = (p : E \to B, f)$ exists by the universal property of the pullback.

$$\begin{array}{ccc} \mathbf{P} & \longrightarrow & \mathbf{P}_1 \\ \downarrow & \lrcorner & \downarrow{\scriptstyle \phi_1} \\ \mathbf{P}_2 & \xrightarrow[\phi_2]{} & \mathbf{M} \end{array}$$

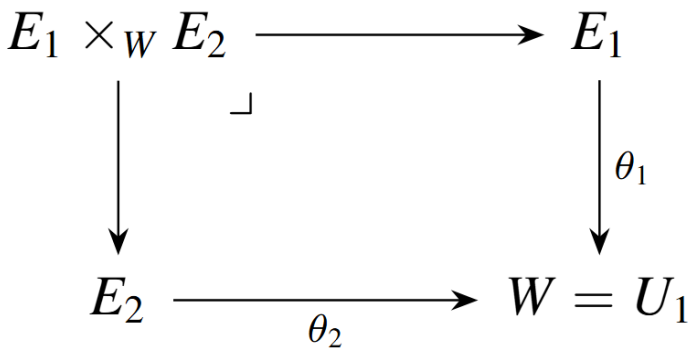
Recover: Cascade Interconnection

Caltech

$\mathcal{C} = \mathsf{Man}, \mathcal{F} = T, E_i = B_i \times U_i, \dot{x}_i = f_i(x_i, u_i).$



Stateless mediator: $\mathbf{M}_W := (! : W \to 1, !)$



$$\theta_1(x_1, u_1) = u_1,$$
$$\theta_2(x_2, u_2) = y_2(x_2).$$

Pullback imposes interconnection:

Composed state-input space: $E \cong B_1 \times B_2 \times U_2$.
Composed dynamics:

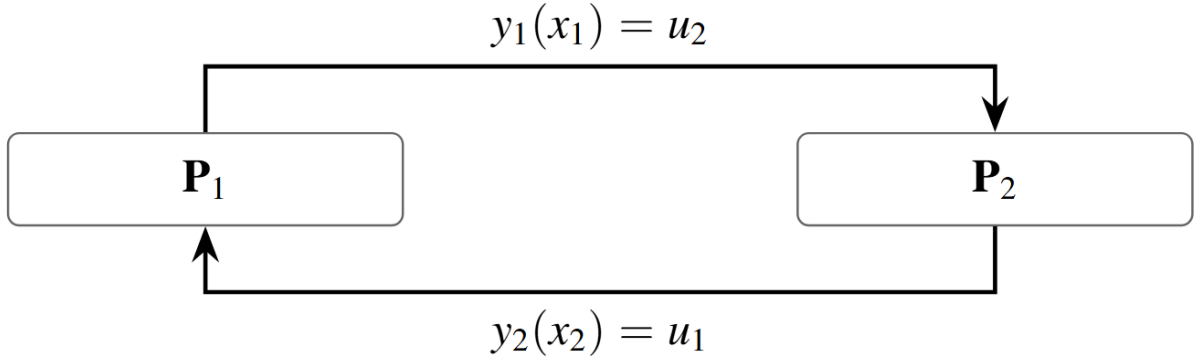
$$\dot{x}_1 = f_1(x_1, y_2(x_2)),$$
$$\dot{x}_2 = f_2(x_2, u_2).$$

Matching a port $u_1 = y_2(x_2)$ leaves a single external input u_2.

Recover: Feedback Interconnection

Caltech

Same systems, with outputs $y_1 : B_1 \to U_2$ and $y_2 : B_2 \to U_1$.



Two-port mediator: $W = U_1 \times U_2$

$$\begin{array}{ccc} \mathbf{P}_1 \times_{\mathbf{M}_W} \mathbf{P}_2 & \longrightarrow & \mathbf{P}_1 \\ \downarrow & \lrcorner & \downarrow{\scriptstyle \theta_1} \\ \mathbf{P}_2 & \xrightarrow[\theta_2]{} & \mathbf{M}_W \end{array}$$

$$\theta_1(x_1, u_1) = (u_1, y_1(x_1)),$$
$$\theta_2(x_2, u_2) = (y_2(x_2), u_2).$$

The composition is autonomous:

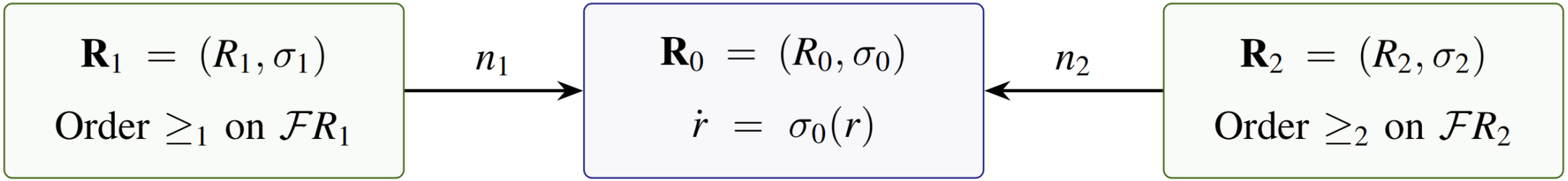
Composed state-input space: $E \cong B_1 \times B_2 \cong B$.
Composed dynamics:

$$\dot{x}_1 = f_1(x_1, y_2(x_2)),$$
$$\dot{x}_2 = f_2(x_2, y_1(x_1)).$$

Composing Comparison Systems

Caltech

Choose autonomous targets $\mathbf{R}_i = (R_i, \sigma_i)$ and **strict** connecting maps.



$$\begin{array}{ccc} \mathbf{Q} := \mathbf{R}_1 \times_{\mathbf{R}_0} \mathbf{R}_2 & \xrightarrow{p_1^R} & \mathbf{R}_1 \\ \downarrow{\scriptstyle p_2^R} & \lrcorner & \downarrow{\scriptstyle n_1} \\ \mathbf{R}_2 & \xrightarrow[n_2]{} & \mathbf{R}_0 \end{array}$$

$$\mathbf{Q} = (R, \sigma), \quad R = R_1 \times_{R_0} R_2.$$

Preorder on Q induced by the projections

For all $X \in \mathsf{C}$ and $\xi, \eta : X \to \mathcal{F}R$ define the order:

$$\xi \geq \eta \iff \mathcal{F}p_i^R \circ \xi \geq_i \mathcal{F}p_i^R \circ \eta, \quad i = 1, 2.$$

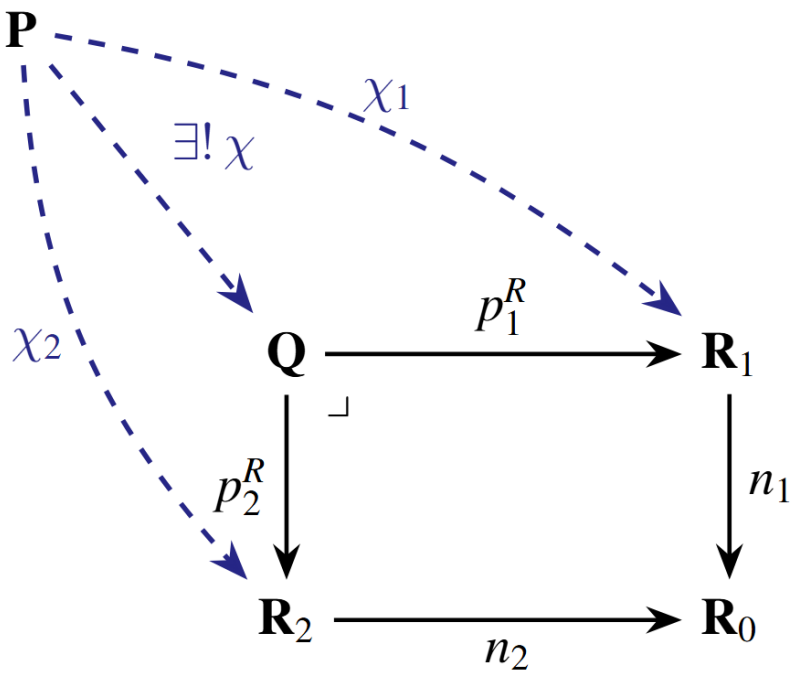
The components may test **different inequalities**: barrier bound ($\geq_1 = \geq_h$), tracking bound ($\geq_2 = \leq_V$).

An autonomous target system **Q** for guarantees on the composed system **P**.

Theorem: Composing Certificates

Caltech

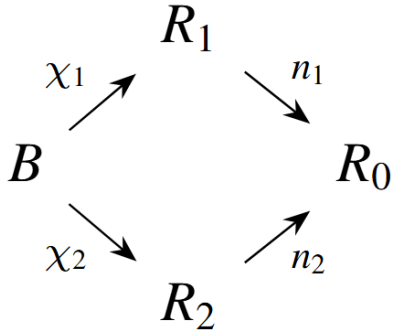
For $\mathbf{P} = \mathbf{P}_1 \times_{\mathbf{M}} \mathbf{P}_2 = (p : E \to B, f)$



Solid: strict system maps.
Dashed: state maps with lax dynamics.

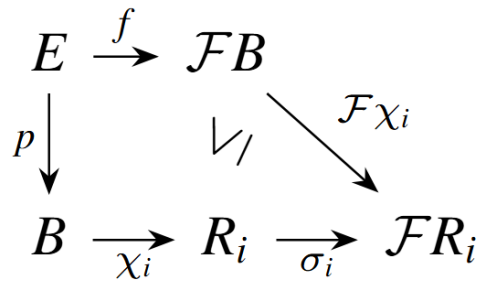
Admissibility: The inequalities hold on all of E, so $\mathcal{A}_\chi \cong E$.

Suppose certificates $\chi_i : B \to R_i$ satisfying:



$n_1 \circ \chi_1 = n_2 \circ \chi_2$

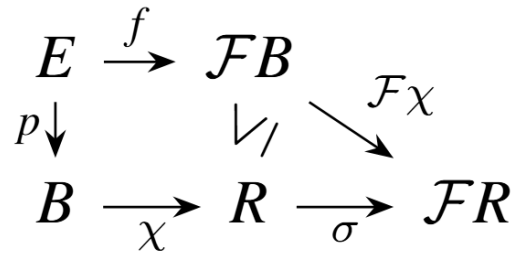
&



$\mathcal{F}\chi_i \circ f \geq_i \sigma_i \circ \chi_i \circ p$

Theorem (Main Composition Result)

There is a unique $\chi : B \to R$ satisfying



$\mathcal{F}\chi \circ f \geq \sigma \circ \chi \circ p$

Corollary: Safety of Composed System

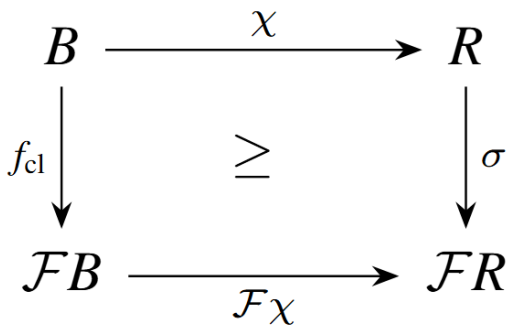
Caltech

Assume: The target R is posetal with distinguished zero 0_R and the barrier comparison property.

The composed barrier morphism

For a feedback section $k : B \to E$,

$$p \circ k = \mathrm{id}_B, \qquad f_{\mathrm{cl}} := f \circ k.$$



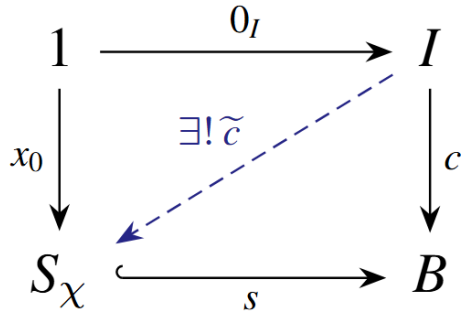
$$\boxed{\mathcal{F}\chi \circ f_{\mathrm{cl}} \geq \sigma \circ \chi.}$$

The safe subobject is forward invariant

Assume the safe subobject exists:

$$\boxed{S_\chi := \mathrm{Eq}^{\leq}(0_B, \chi).}$$

For solutions c with $c \circ 0_I = s \circ x_0$:



Corollary (Safety by certificate composition)

Under the preceding theorem, any feedback section k renders S_χ forward invariant for f_{cl}.

Layered Control Architectures - Revisited

Caltech

The layered safety architecture

Reduced-order model (RoM)

$$\dot{z} = \nu, \qquad h(z(t)) \geq 0$$

Top interface

z

Safety filter

$$\kappa_{\text{safe}}(z) = \underset{\nu \in \mathbb{R}^n}{\operatorname{argmin}} \frac{1}{2}\|\nu - \nu_{\text{des}}(z)\|^2$$

$$\text{s.t.} \quad Dh(z)\nu \geq -\alpha h(z)$$

Bottom interface

$v = \kappa_{\text{safe}}(z)$

Full-order model (FoM)

$$D(q)\ddot{q} + C(q,\dot{q}) = Bu$$

$$u = K(q,\dot{q},v), \quad h(q(t)) \geq 0$$

$z = q$

The categorical representation

$$\mathbf{P}_{\text{RoM}} = \left(Z \times \mathbb{R}^n_\nu \xrightarrow{p_z} Z, f_{\text{R}}\right)$$

$$\dot{z} = f_{\text{R}}(z,\nu) = \nu$$

ϕ_R

Safety-filter mediator

$$\mathbf{M}_{\text{sf}} = \left(Z \times \mathbb{R}^n_\nu \times \mathbb{R}^n_v \xrightarrow{p_0} Z, f_{\text{sf}}\right)$$

$$\dot{z} = f_{\text{sf}}(z,\nu,v) = \nu$$

ϕ_F

$$\mathbf{P}_{\text{FoM}} = \left(M \times \mathbb{R}^n_v \xrightarrow{p_M} M, f_{\text{F}}\right)$$

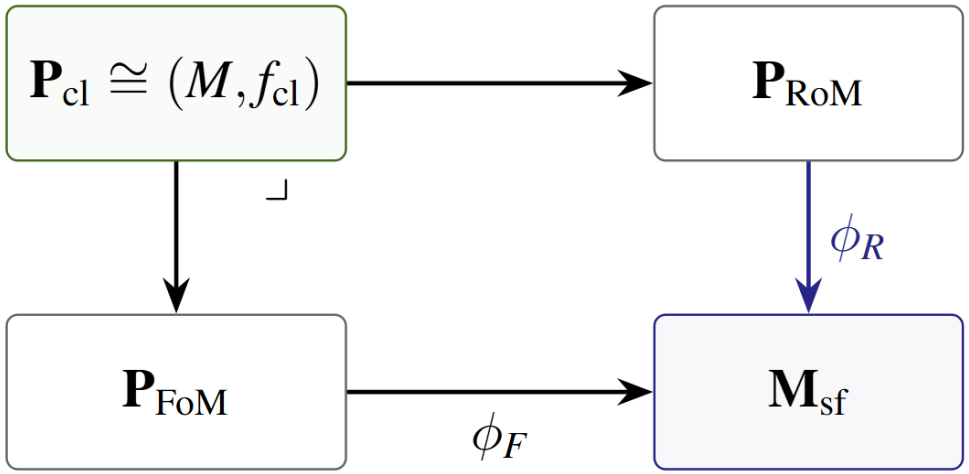
$$\dot{x} = f_{\text{F}}(x,v), \qquad u = K(q,\dot{q},v)$$

Here $x = (q,\dot{q}) \in M \subset TQ$ and $Z = Q \subset \mathbb{R}^n$

Composition

Caltech

Composition is the pullback



Composition: **closed-loop system**:

$$\mathbf{P}_{\mathrm{cl}} = \mathbf{P}_{\mathrm{RoM}} \times_{\mathbf{M}_{\mathrm{sf}}} \mathbf{P}_{\mathrm{FoM}} \cong (M, f_{\mathrm{cl}}),$$
$$\dot{x} = f_{\mathrm{cl}}(x) = f_{\mathrm{F}}(x, \kappa_{\mathrm{safe}}(q)).$$

The categorical representation

$$\mathbf{P}_{\mathrm{RoM}} = \left(Z \times \mathbb{R}^n_\nu \xrightarrow{p_z} Z, f_{\mathrm{R}}\right)$$
$$\dot{z} = f_{\mathrm{R}}(z, \nu) = \nu$$

ϕ_R

Safety-filter mediator

$$\mathbf{M}_{\mathrm{sf}} = \left(Z \times \mathbb{R}^n_\nu \times \mathbb{R}^n_v \xrightarrow{p_0} Z, f_{\mathrm{sf}}\right)$$
$$\dot{z} = f_{\mathrm{sf}}(z, \nu, v) = \nu$$

ϕ_F

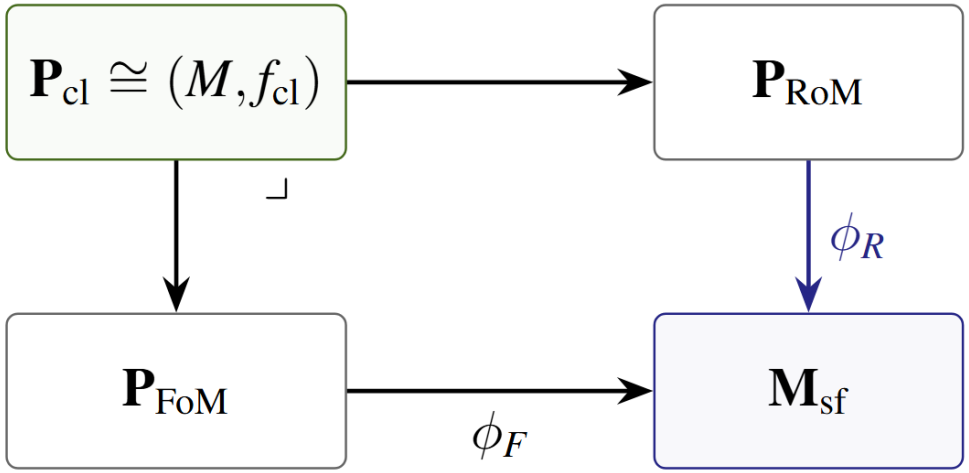
$$\mathbf{P}_{\mathrm{FoM}} = \left(M \times \mathbb{R}^n_v \xrightarrow{p_M} M, f_{\mathrm{F}}\right)$$
$$\dot{x} = f_{\mathrm{F}}(x, v), \qquad u = K(q, \dot{q}, v)$$

How do we guarantee safety of the closed-loop system $\mathbf{P}_{\mathrm{cl}}$?

Composition

Caltech

Composition is the pullback



Composition: **closed-loop system**:

$$\mathbf{P}_{\text{cl}} = \mathbf{P}_{\text{RoM}} \times_{\mathbf{M}_{\text{sf}}} \mathbf{P}_{\text{FoM}} \cong (M, f_{\text{cl}}),$$
$$\dot{x} = f_{\text{cl}}(x) = f_{\text{F}}(x, \kappa_{\text{safe}}(q)).$$

RoM interface: the QP graph

$$\phi_R^B(z) = z,$$
$$\phi_R^E(z, \nu) = (z, \nu, \kappa_{\text{safe}}(z)).$$

FoM interface: measured motion

$$\phi_F^B(q, \dot{q}) = q,$$
$$\phi_F^E(q, \dot{q}, v) = (q, \dot{q}, v).$$

Pullback: enforces

$$z = q, \qquad \nu = \dot{q}, \qquad v = \kappa_{\text{safe}}(q)$$

How do we guarantee safety of the closed-loop system $\mathbf{P}_{\text{cl}}$?

Certificates for RoM and FoM

$$\begin{aligned} \mathbf{P}_{\mathrm{RoM}} &= (Z \times \mathbb{R}^n_\nu \xrightarrow{p_z} Z, f_{\mathrm{R}}) \\ \dot{z} &= f_{\mathrm{R}}(z,\nu) = \nu \end{aligned}$$

$$\begin{aligned} \mathbf{P}_{\mathrm{FoM}} &= (M \times \mathbb{R}^n_v \xrightarrow{p_M} M, f_{\mathrm{F}}) \\ \dot{x} &= f_{\mathrm{F}}(x,v), \quad u = K(q,\dot{q},v) \end{aligned}$$

$\mathbf{P}_{\mathrm{RoM}}$: RoM safety and barrier certificate

$$\begin{array}{ccc} Z & \xrightarrow{h} & \mathbb{R} \\ {\scriptstyle \kappa_{\mathrm{safe}}}\downarrow & \geq & \downarrow{\scriptstyle -\alpha(\cdot)} \\ TZ & \xrightarrow[Th]{} & T\mathbb{R} \end{array}$$

RoM CBF: $Dh(z)\kappa_{\mathrm{safe}}(z) \geq -\alpha h(z),$

$$b(x) := h(q) \implies \boxed{\dot{b} \geq -\alpha b - \gamma V.}$$

$\mathbf{P}_{\mathrm{FoM}}$: Full-order tracking certificate

$$\begin{array}{ccc} M & \xrightarrow{V} & \mathbb{R} \\ {\scriptstyle f_{\mathrm{cl}}}\downarrow & \leq & \downarrow{\scriptstyle -\lambda(\cdot)} \\ TM & \xrightarrow[TV]{} & T\mathbb{R} \end{array}$$

Tracking error: $e(x) := \dot{q} - \kappa_{\mathrm{safe}}(q)$

Tracking function : $V(x) \geq m_V \|e(x)\|$

$$\dot{V}(x) = DV(x) f_{\mathrm{cl}}(x) \implies \boxed{\dot{V} \leq -\lambda V.}$$

Tracking error dynamics

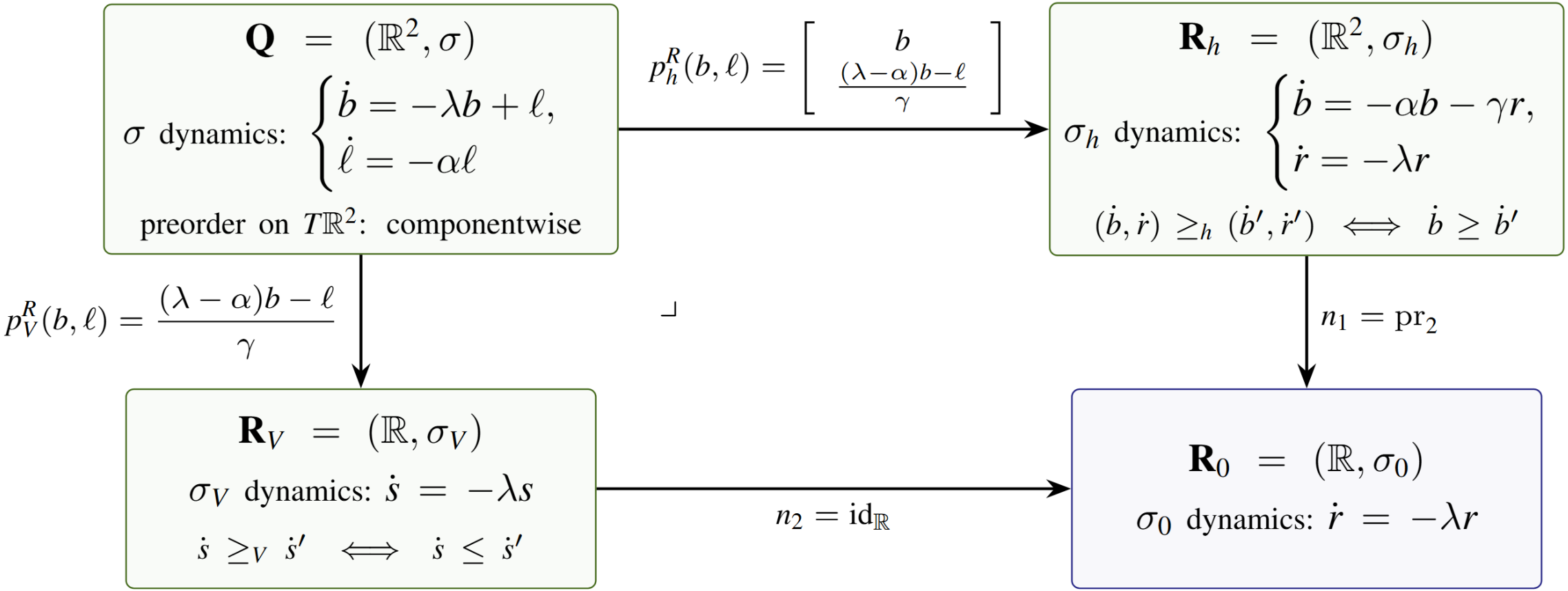
$\dot{h} = Dh(q)[\kappa_{\mathrm{safe}}(q) + e(x)] \geq -\alpha h - \gamma V$, where V bounds the tracking-induced loss of safety margin.

Set $\gamma := C_h/m_V$ for $C_h \geq \|Dh(z)\|$. Assume $C_h, m_V > 0$ and $\lambda > \alpha \geq 0$.

Comparison System

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The comparison target system is the pullback:

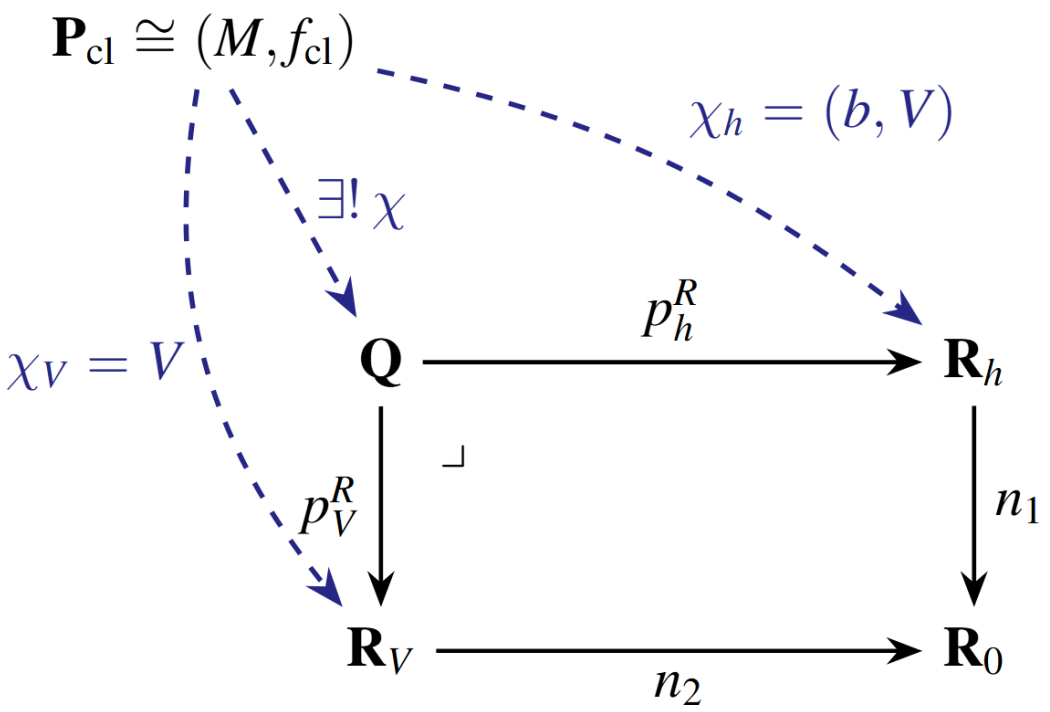


Q is the pullback with coordinates (b, ℓ), where ℓ is the remaining safety margin.

Induced Certificate on Composition

Caltech

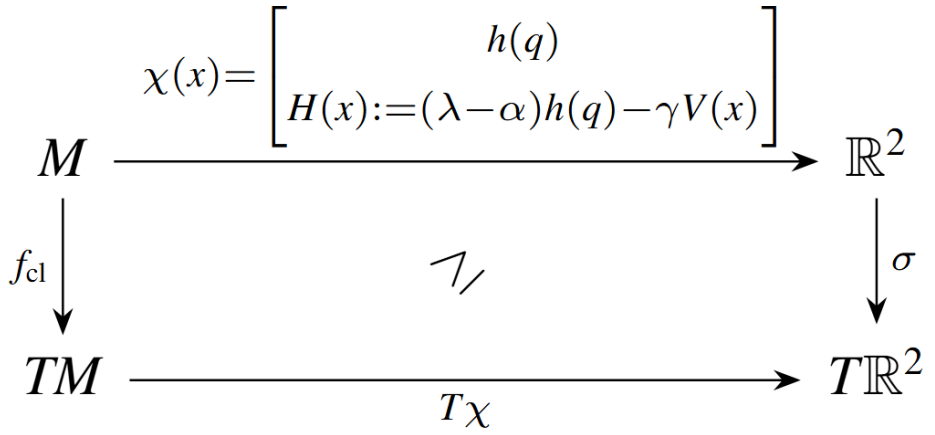
Both certificates have the same autonomous source:



Solid: strict system maps.
Dashed: state maps with lax dynamics.

Theorem: ∃! induced certificate

There exists a unique $\chi = (h, H) : M \to \mathbb{R}^2$ with



$$\boxed{T\chi \circ f_{\mathrm{cl}} \geq \sigma \circ \chi.}$$

Proof: by Main Composition Theorem

The component certificates χ_h and χ_V satisfy:

$$n_1 \circ \chi_h = n_2 \circ \chi_V, \qquad \begin{aligned} T\chi_h \circ f_{\mathrm{cl}} &\geq_h \sigma_h \circ \chi_h, \\ T\chi_V \circ f_{\mathrm{cl}} &\geq_V \sigma_V \circ \chi_V. \end{aligned}$$

Safety of Composition

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The safe set on the robot

- $\chi : M \to \mathbb{R}^2$ defines the safe set S_χ
- Since $\chi(x) = \big(h(q), H(x)\big)$,

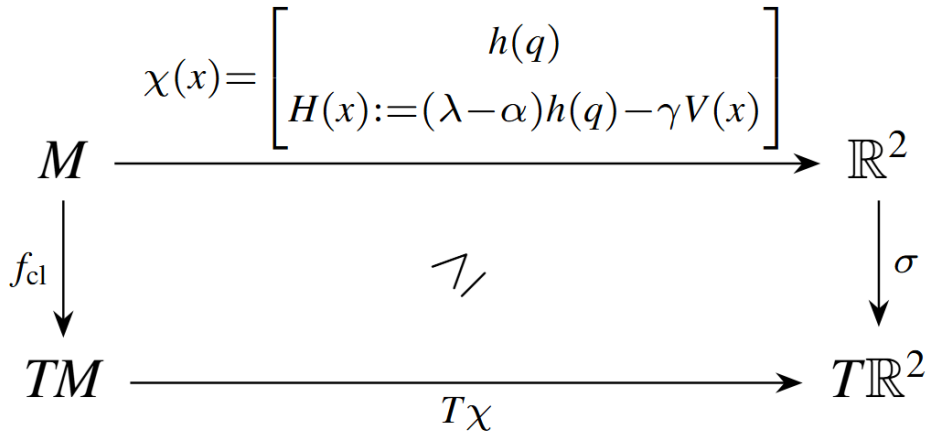
$$\begin{aligned} x \in S_\chi &\iff \chi(x) \geq 0_R \\ &\iff \begin{matrix} h(q) \geq 0, \\ H(x) = (\lambda - \alpha)h(q) - \gamma V(x) \geq 0 \end{matrix} \end{aligned}$$

- $H \geq 0$ implies $h \geq 0$ (if $\lambda > \alpha$ and $V \geq 0$):

$$\boxed{\begin{aligned} S_\chi &= \{x \in M : H(x) \geq 0\} \\ &= \{x \in M : (\lambda - \alpha)h(q) \geq \gamma V(x)\}. \end{aligned}}$$

Theorem: ∃! induced certificate

There exists a unique $\chi = (h, H) : M \to \mathbb{R}^2$ with



$$\boxed{T\chi \circ f_{\text{cl}} \geq \sigma \circ \chi.}$$

Corollary: safety of the closed-loop system $\mathbf{P}_{\text{cl}} \cong (M, f_{\text{cl}})$

Every solution $x : I \to M$ of f_{cl}, $\dot{x}(t) = f_{\text{cl}}(x(t))$, starting in S_χ guarantees system level safety:

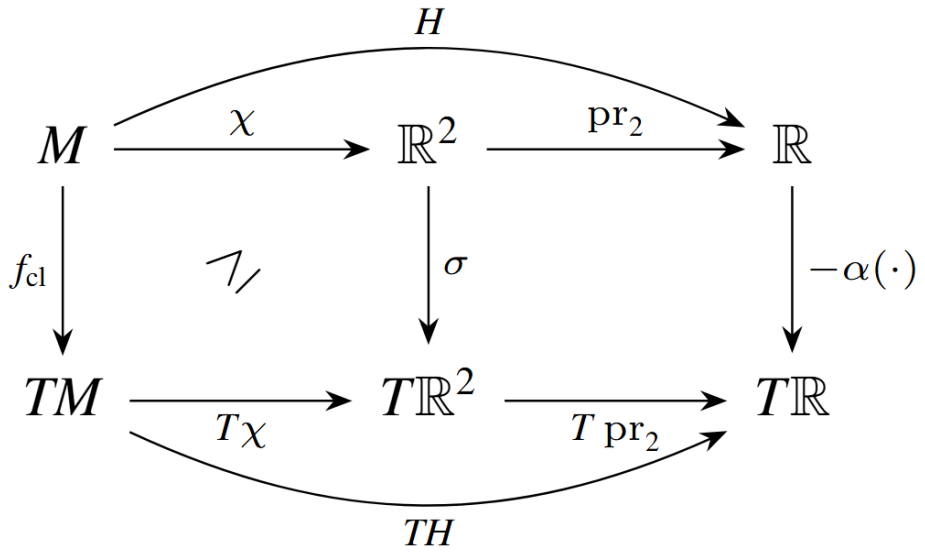
$$x(0) \in S_\chi \implies x(t) \in S_\chi \implies H(x(t)) \geq 0 \implies h(q(t)) \geq 0, \qquad t \in I.$$

Recover Classic Result

Caltech

The classical barrier by pasting

The morphism $\chi = (h, H) : M \to \mathbb{R}^2$ satisfies:



Recover classic result: $\boxed{\dot{H} = TH \circ f_{\text{cl}} \geq -\alpha H}$.

Theorem (classic)

If $\lambda > \alpha$ then $S_\chi \hookrightarrow \mathcal{C} = \{(q, \dot{q}) : h(q) \geq 0\}$ so:

$$\boxed{H(x_0) \geq 0 \implies h(q(t)) \geq 0 \quad \forall t \geq 0.}$$

Where the conditions come from

1. Pullback determines the certificate

The pullback projections satisfy

$$p_h^R \circ \chi = (h, V), \qquad p_V^R \circ \chi = V,$$

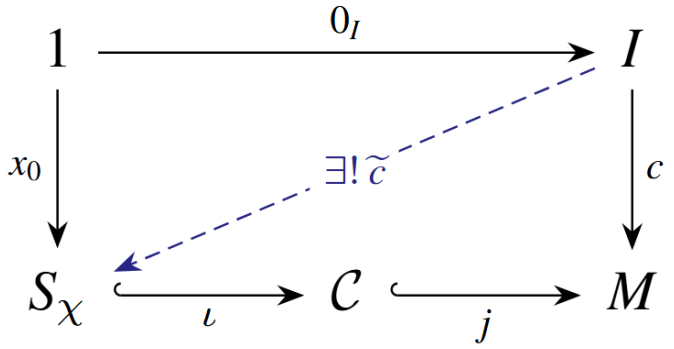
and determine the unique map

$$\boxed{\chi = \big(h, (\lambda - \alpha)h - \gamma V\big) =: (h, H).}$$

2. Parameter choice implies safety

Safety corollary implies S_χ is forward invariant.

$\lambda > \alpha \implies$



Extensions: General RoMs

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The layered safety architecture

Reduced-order model (RoM)

$$\dot{z} = f(z) + g(z)\nu, \qquad h(z(t)) \geq 0$$

Top interface

z

Safety filter

$$\kappa_{\text{safe}}(z) = \underset{\nu \in \mathbb{R}^n}{\operatorname{argmin}} \frac{1}{2}\|\nu - \nu_{\text{des}}(z)\|^2$$

$$\text{s.t.} \quad \dot{h}(z,\nu) \geq -\alpha h(z)$$

Bottom interface

$\nu = \kappa_{\text{safe}}(\Pi(x))$

Full-order model (FoM)

$$D(q)\ddot{q} + C(q,\dot{q}) = Bu$$

$$u = K(q,\dot{q},\nu), \quad h(\Pi(x(t))) \geq 0$$

$$z = \Pi(x)$$
$$\nu = \Psi(x)$$

More expressive reduced-order model

$$\mathbf{P}_{\text{RoM}} = \left(Z \times U \xrightarrow{p_z} Z, f_{\text{R}}\right)$$
$$\dot{z} = f_{\text{R}}(z,\nu) = f(z) + g(z)\nu$$

Requires the categorical condition:

$$\begin{array}{ccc} M \times U & \xrightarrow{(\Pi,\Psi)} & Z \times U \\ \downarrow{\scriptstyle f_{\text{F}}} & & \downarrow{\scriptstyle f_{\text{R}}} \\ TM & \xrightarrow[T\Pi]{} & TZ \end{array}$$

$$D\Pi(x) f_{\text{F}}(x,\nu) = f_{\text{R}}(\Pi(x), \Psi(x))$$

Then same framework applies with:

$$b(x) = h(\Pi(x)),$$
$$e(x) = \Psi(x) - \kappa_{\text{safe}}(\Pi(x)).$$

Category theory enables the extension to a wide class of layered architectures.

Safe Humanoids

Reduced-order model (RoM)

$$\dot{z} = f(z) + g(z)\nu, \qquad h(z(t)) \geq 0$$

Top interface

z

Safety filter

$$\kappa_{\text{safe}}(z) = \underset{\nu \in \mathbb{R}^n}{\operatorname{argmin}} \frac{1}{2}\|\nu - \nu_{\text{des}}(z)\|^2$$

$$\text{s.t.} \quad \dot{h}(z,\nu) \geq -\alpha h(z)$$

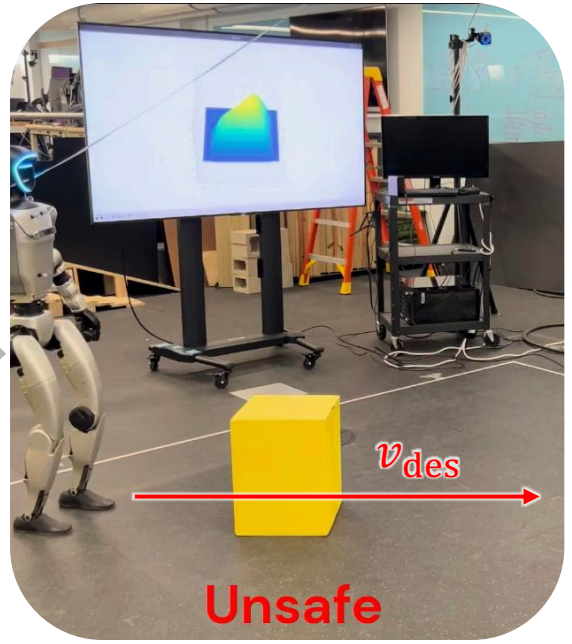
Bottom interface

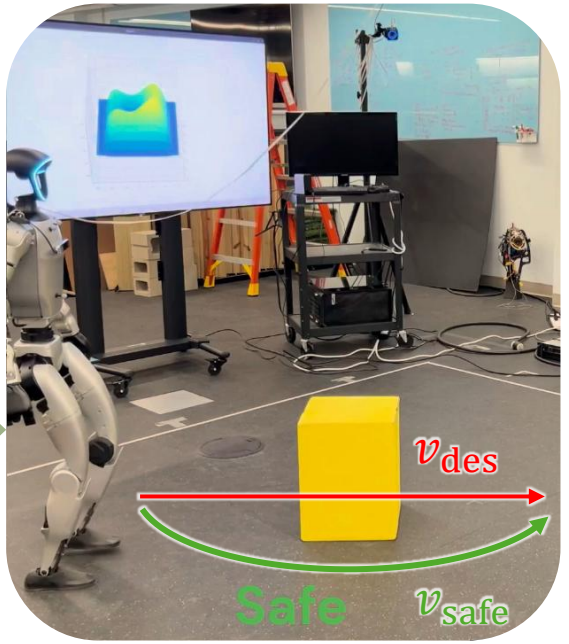
$\nu = \kappa_{\text{safe}}(\Pi(x))$

Full-order model (FoM)

$$D(q)\ddot{q} + C(q,\dot{q}) = Bu$$

$$u = K(q,\dot{q},\nu), \quad h(\Pi(x(t))) \geq 0$$

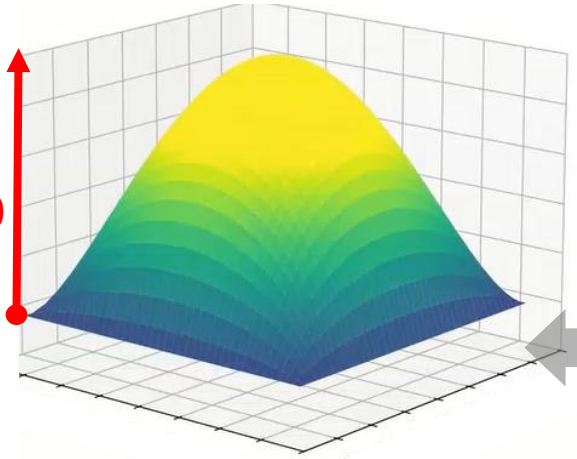


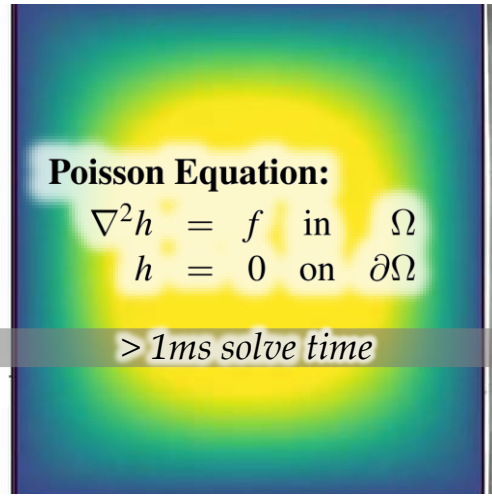


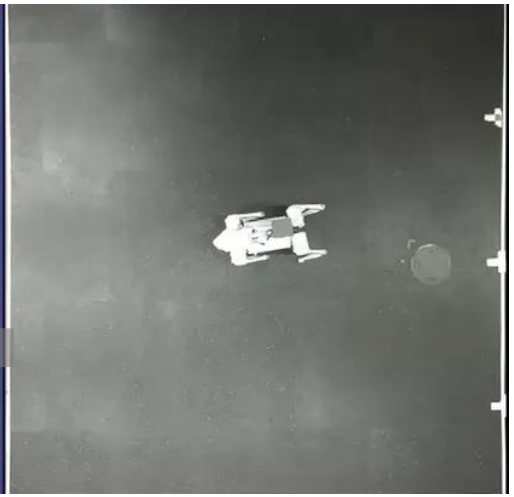
Bahati, Bena, AA, RSS 2025

Safe Humanoids

Caltech







RoM

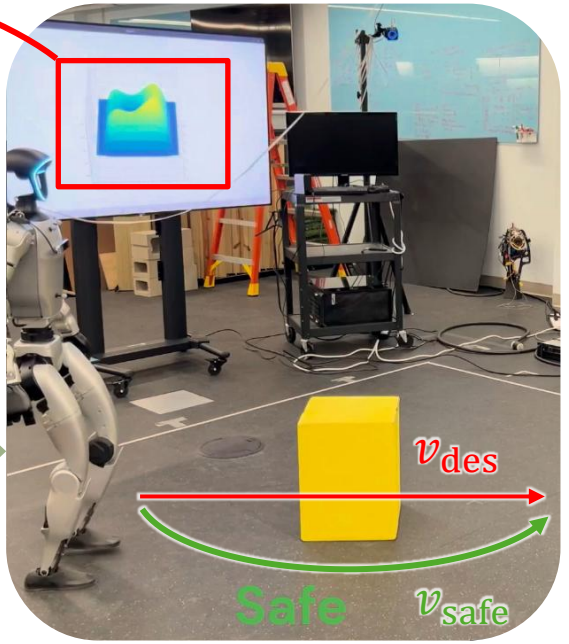
Safety filter

$$\kappa_{\text{safe}}(z) = \underset{\nu \in \mathbb{R}^n}{\text{argmin}} \frac{1}{2}\|\nu - \nu_{\text{des}}(z)\|^2$$
$$\text{s.t.} \quad \dot{h}(z,\nu) \geq -\alpha h(z)$$

FoM

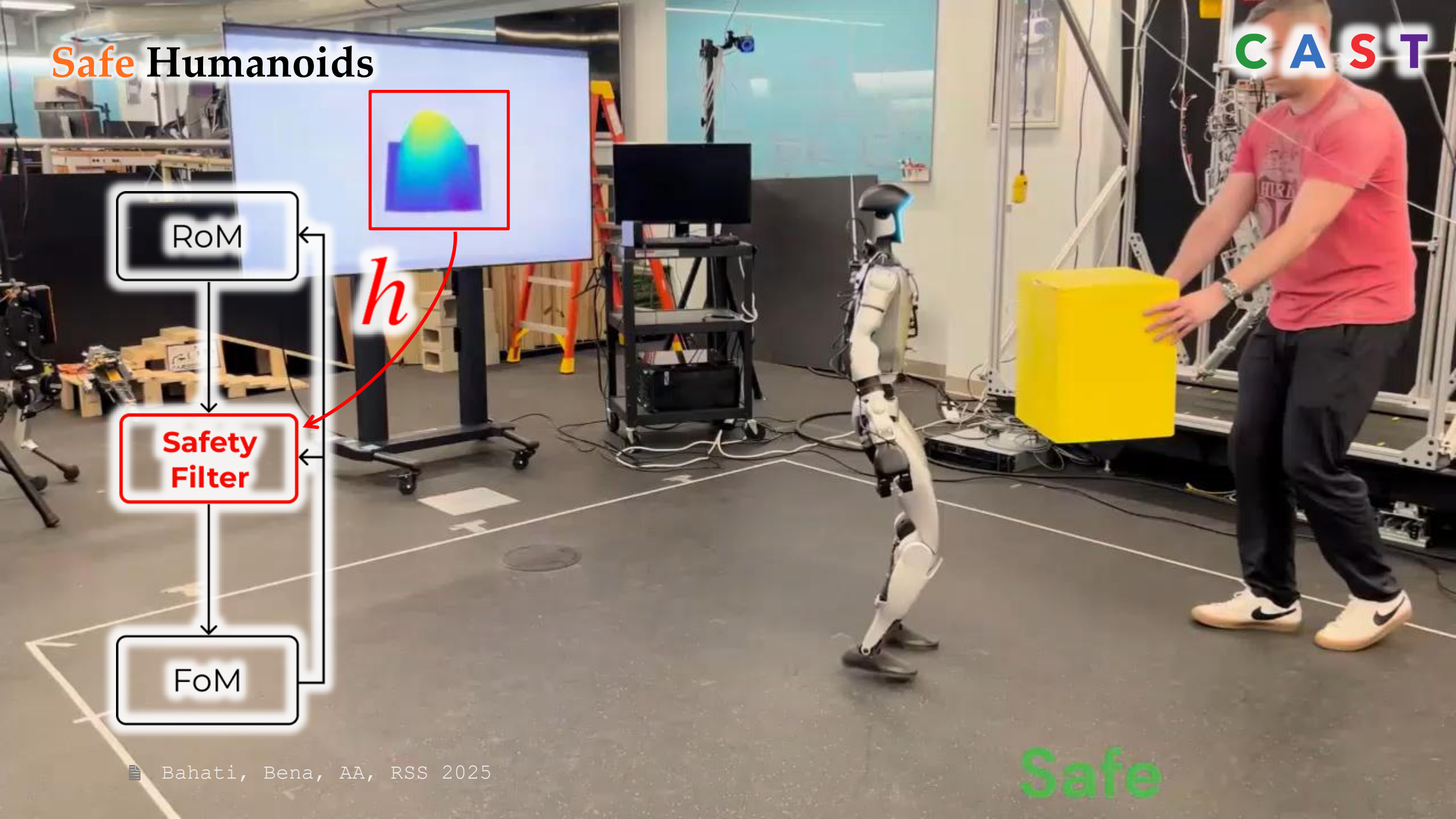
Poisson Safety Function: h

Smooth CBF synthesis from perception



Bahati, Bena, AA, RSS 2025

Safe Humanoids



Bahati, Bena, AA, RSS 2025

Safe Humanoids



Yamaguchi, Bena, Bahati, Ames, RAL 2026

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Guardrails on the VISTA F-16

- **Implemented by 3Laws at Edwards AFB Test Pilot School: September 09-20, 2024**
- Completed all 14 test points including pilot assault on safety limits – *100% success rate*
- Constraints (geofences, altitudes, etc) were altered in-flight from ground station



Pilot authority is modified only when near and approaching a constraint limit. No impact on flying characteristics otherwise—*minimally invasive*.



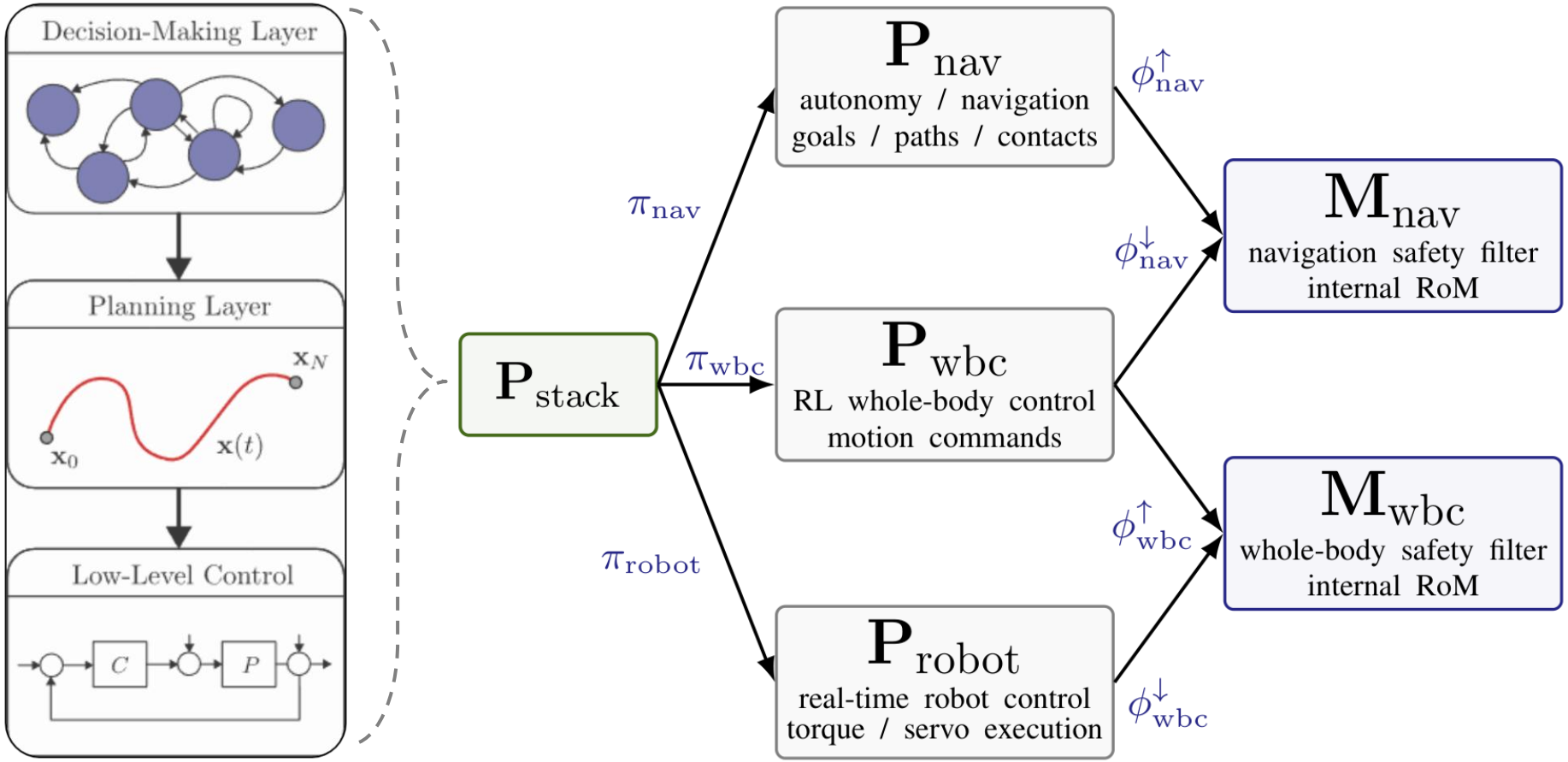
3Laws Robotics

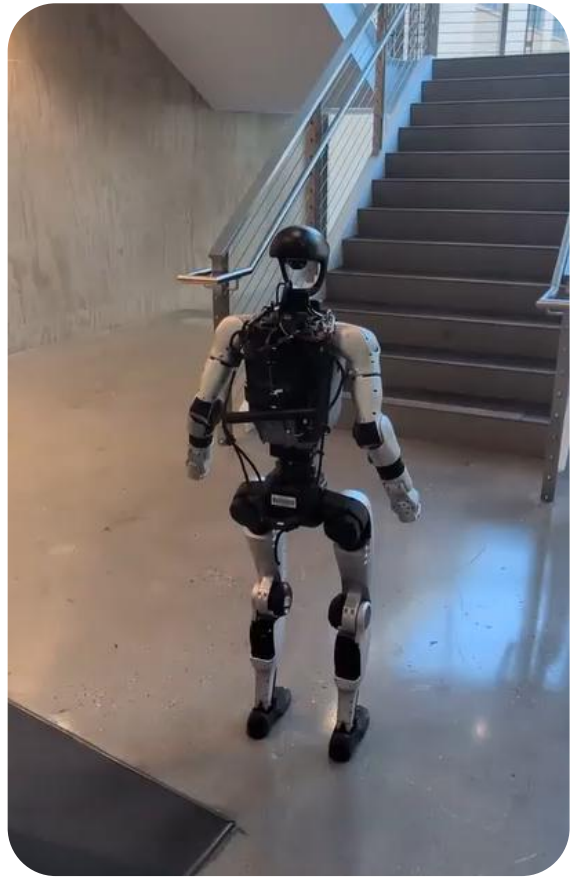

GEOFENCING +
ALTITUDE LIMITING
3Laws
Robotics

Categorical Systems Theory – Needed to Scale

Caltech

$$\mathbf{P}_{\text{stack}} \cong \lim\left(\mathbf{P}_{\text{nav}} \to \mathbf{M}_{\text{nav}} \leftarrow \mathbf{P}_{\text{wbc}} \to \mathbf{M}_{\text{wbc}} \leftarrow \mathbf{P}_{\text{robot}}\right)$$





Goal: compositional guarantees for the full autonomy stack

Goal: Category Theory on Robots

